

# Dominic M. Bowman

PhD, MSci, FRAS, MInstP, FHEA

**Reader in Astrophysics and Royal Society University Research Fellow**

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## I. Professional Profile

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Holder of a Readership faculty position, a Royal Society University Research Fellowship, and PI of an ERC/UKRI Frontier Research Grant (*SYMPHONY*) at [Newcastle University](#). Extensive expertise in the extraction and analysis of photometric and spectroscopic data from space- and ground-based telescopes, and forward asteroseismic modelling of pulsating stars, which yields tight constraints on their interior physics such as rotation, mixing, magnetism, and angular momentum transport. Passionate and actively involved in developing teaching at the BSc, MSc, and PhD level, as well as mentoring, advocacy, and outreach activities for all ages and backgrounds.

Completed a PhD in Astronomy at the [University of Lancashire](#) under the supervision of Prof. Donald Kurtz, and PhD thesis was published as a [Springer monograph](#). International move to [KU Leuven](#) as a postdoctoral researcher, and later awarded a competitive [FWO](#) research fellowship. Over 120 peer-reviewed publications, with dozens of high-impact first-author papers including in [Nature Astronomy](#). Winner of several prestigious prizes for research excellence and an upwards career trajectory including: [Springer Thesis Prize](#) (2017); [KU Leuven Research Council Award in Science, Engineering and Technology](#) (2020); [Henri Vanderlinden Prize](#) of the Flemish Academy (2022); and [George Darwin Lectureship](#) of the Royal Astronomical Society (2023).

## II. Education

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### PhD in Astronomy

Oct 2013 – Nov 2016

Thesis title of '*Amplitude modulation and energy conservation of pulsation modes in delta Scuti stars*', awarded outright (no corrections) on 21 November 2016 by the University of Lancashire, with supervisor of Prof. Donald Kurtz, and funded by the UK Science and Technology Facilities Council (STFC).

### MSci in Physics and Astrophysics

Sep 2009 – Jun 2013

First-class integrated Master (BSc + MSc) in Science (MSci) degree with honours in Physics and Astrophysics from the University of Birmingham, with an award date of 8 July 2013.

## III. Employment

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### Reader in Astrophysics

1 Sep 2023 – date

School of Mathematics, Statistics and Physics, Newcastle University, Newcastle upon Tyne, NE1 7RU, UK.

### Guest Professor

1 Sep 2023 – 31 Aug 2026

Institute of Astronomy, KU Leuven, Celestijnenlaan 200D, 3001 Leuven, Belgium.

### FWO Senior Postdoctoral Fellowship

1 Nov 2020 – 31 Aug 2023

Institute of Astronomy, KU Leuven, Celestijnenlaan 200D, 3001 Leuven, Belgium. Independent research fellowship funded by Fonds Wetenschappelijk Onderzoek (FWO) Vlaanderen [PI Bowman; grant number: 1286521N].

### Postdoctoral Research Associate

1 Feb 2017 – 31 Oct 2020

Institute of Astronomy, KU Leuven, Celestijnenlaan 200D, 3001 Leuven, Belgium. Funded by the European Union's Horizon 2020 research and innovation programme [ERC-AdG; PI Aerts; grant number: 670519].

### Lecturer in Astronomy

19 Sep 2016 – 13 Jan 2017

Jeremiah Horrocks Institute, University of Lancashire, Preston, PR1 2HE, UK.

## IV. Scientific Prizes, Awards and Competitive Grant Funding

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Multiple prestigious prizes for scientific excellence and having an upwards career trajectory. A funding portfolio as PI from successful competitive applications to date that exceeds **€5 M (£4.3 M)**, which includes international research councils and charities including the Royal Society, UKRI, ERC, FWO, and several universities.

### Scientific Prizes and Awards:

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**George Darwin Lectureship, RAS** 2023

Awarded the 2023 [George Darwin Lectureship](#) of the Royal Astronomical Society (RAS) for being an authoritative and engaging researcher in astronomy.

**Henri Vanderlinden Prize, Royal Flemish Academy** 2022

Prestigious and competitive prize for an important original work in the field of astronomy from the [Koninklijke Vlaamse Academie van België](#) (KVAB) voor Wetenschappen en Kunsten, which included a cash prize.

**Research Council Award (POR), KU Leuven** 2020

Prestigious and highly-competitive prize from KU Leuven's [Research Council](#) awarded to only one postdoc in STEM for research excellence and an upwards career trajectory, which included **€20,000** of research funding.

**Springer Thesis Award** 2017

PhD thesis selected to be published in the [Springer thesis series](#) in 2017 for 'Outstanding PhD Research', which included a cash prize.

### Successful Competitive Grant Applications:

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**Royal Society University Research Fellowship** Dec 2023 – Nov 2031

Awarded a coveted Royal Society [University Research Fellowship](#) (URF; **£1,260,000**) for the ECLIPSE project focussed on massive stars in eclipsing binaries [grant number: URF\R1\231631].

**ERC-StG-2023** Sep 2023

Invited to sign grant agreement for 2023 call of [ERC starting grant](#) (**€1,500,000**). Unable to sign ERC grant agreement because UK was at that time not formally associated to ERC Horizon framework, so declined.

**ERC-StG-2022 / UKRI Frontier Research Grant** Oct 2023 – Sep 2028

Invited to sign grant agreement for 2022 call of ERC starting grant for the SYMPHONY project focused on blue supergiant asteroeismology. Unable to sign ERC grant agreement because UK was at that time not formally associated to ERC Horizon framework. Awarded equal funding under [UKRI's Horizon Guarantee Scheme](#) to implement the [SYMPHONY](#) project at a UK host institution (**£1,270,000**) as a UKRI Frontier Research Grant [grant number: EP/Y031059/1].

**FWO Postdoctoral Fellowship** Nov 2020 – Aug 2023

Awarded a senior postdoctoral fellowship of [Fonds Wetenschappelijk Onderzoek](#) (FWO) Vlaanderen (**€300,000**) for the TESSERACT asteroeismology project [grant number: 1286521N].

**FWO Long Stay Abroad Grant** Oct – Dec 2021

Awarded a competitive FWO grant for a long research stay abroad to cover all costs for an invited visit to KITP, California, USA for 3 months in 2021, for a total of approximately **€6000** [grant number: V411621N].

**Conference Organisation** Sep 2015

Successful funding applications of **£3000** from the Royal Astronomical Society (RAS) and **£7000** from University of Lancashire for organisation of the STARS2016 conference together with Dr. Daniel Holdsworth.

### Small Travel Grants

Numerous travel bursary applications from funding bodies for attending conferences, which include STFC and RAS in the UK, CNRS in France, and FWO in Belgium, as well as travel costs for obtaining competitive telescope time in visitor mode (e.g. ESO), with a combined total of approximately **£15,000**.

## V. Conference Organisation

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Three times (co-)chair of an SOC, twice chair of an LOC, and four times a member of an SOC for large international conferences and workshops.

**TASC9/KASC16**, *ISTA, Austria* 7–11 Jul 2025

SOC member for the annual asteroseismology conference, which had 200+ participants.

**Stellar Hydro Days VI**, *University of Victoria, Canada* 12–16 May 2025

SOC member for a workshop on bridging MHD simulators and observers, which had 20+ participants.

**TASC8/KASC15**, *Porto, Portugal* 15–19 Jul 2024

SOC member for the annual asteroseismology conference, which had 200+ participants.

**EAS 2023**, *Kraków, Poland* 10–14 Jul 2023

SOC member of the BRITE/MOBSTER symposium entitled ‘From stellar variability to stellar structure and evolution’ at the EAS 2023 meeting, which had 100+ participants.

**TASC6/KASC13**, *Leuven, Belgium* 11–15 Jul 2022

Chair of the LOC for the annual asteroseismology conference, which had a budget of €70,000, and 200 in-person and 100+ online participants. Postponed from 2020 to 2022 because of the COVID-19 pandemic.

**EAS 2021**, *Virtual (hosted by Leiden University, the Netherlands)* 28 Jun – 2 Jul 2021

Co-Chair of the SOC of the symposium titled ‘Massive stars: birth, rotation, and chemical evolution’ at the EAS 2021 meeting, which had 100+ participants.

**MOBSTER-1**, *Virtual (hosted by University of Delaware, USA)* 13–17 Jul 2020

Co-Chair of the SOC for the virtual MOBSTER-1 conference, which had 170+ participants.

**EAS 2020**, *Virtual (hosted by Leiden University, the Netherlands)* 29 Jun – 3 Jul 2020

Chair of the SOC of the session titled ‘New insights of angular momentum transport in stellar interiors’ held on 1 June during the EAS 2020 meeting, which had 100+ participants.

**STARS2016**, *Windermere, UK* 11–15 Sep 2016

Chair of the LOC for the conference celebrating the career of Prof. Donald Kurtz, which had 75 participants and a budget of £40,000. Successful grant applications included £3000 from the RAS and £7000 from the University of Lancashire.

## VI. Personal Training

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A strong track record and dedicated to continue undertaking CPD training and personal growth in all aspects of an academic career.

**Faculty Training**, *Newcastle University* Sep 2023 – date  
Continuing professional development (CPD) training as a member of faculty, which includes seminars and workshops on EDI, GDPR, project management, student welfare and supervision.

**Leadership Effectiveness**, *Royal Society, London* 3–5 June 2026  
Three-day workshop on leadership hosted by the business school of Imperial College London and the Royal Society.

**Future Leaders: 64 Million Artists**, *Newcastle University* May – Oct 2025  
Several full-day sessions on leadership, project and personnel management for academia and its stakeholders.

**URF Mentoring and Networking**, *Royal Society, London* 13–14 Nov 2024  
Two-day workshop on networking, mentoring and how to manage a successful academic career.

**Bullying and Harassment in Astronomy**, *Virtual, (hosted by RAS)* 17 May 2024  
Workshop on the results of the RAS's bullying and harassment impact survey and improved best practices.

**Decolonising the Curriculum**, *Newcastle University* 8 Apr 2024  
Seminar and discussion session led by Prof. Nira Chamberlain OBE on inclusive education.

**PhD Student Supervision**, *Northumbria University* 8 Dec 2023  
Workshop from higher education and early-career groups of IOP on best practices of PhD supervision.

**Anti-Racism in Astronomy and Geophysics**, *Virtual, (hosted by RAS)* 12 Aug 2021  
Seminars and training sessions on best practices for anti-racism in academia.

**Sex and Gender Dimensions in Frontier Research**, *Virtual (hosted by ERCEA)* 16 Nov 2020  
Seminars and training sessions on diversity initiatives in academia.

**Voice of the Future**, *Westminster, London, UK* 15 Mar 2017  
Nominated an ECR representative of the RAS to attend this meeting on bridging scientists and UK politicians.

**STFC Careers Event**, *Institute of Physics, London* 21 Oct 2015  
Seminars and training sessions on career planning and development.

**Media Training for Outreach**, *Royal Society, London* 7 Oct 2015  
Training sessions on various media-related aspects of outreach activities.

## VII. International Responsibilities, Committees and Service

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### **UK Astronomy Forum Member, RAS**

Jan 2026 – date

Member and Newcastle University representative of the UK's Royal Astronomical Society (RAS) astronomy forum, which is a cross-institution group to advance UK astronomy.

### **WG Chair, HAYDN space mission**

Jan 2026 – date

Coordinator of the 'Massive Stars' working group for the [HAYDN](#) space mission, which was successful at step 1 and under review at step 2 (as of March 2026) for the M8 call of the European Space Agency (ESA).

### **Member of CDAG, RAS**

May 2024 – date

Member of Committee for Diversity in Astronomy and Geophysics ([CDAG](#)) of the Royal Astronomical Society.

### **Executive Organising Committee Member, IAU WGABS**

Nov 2022 – date

Executive [organising committee member](#) for IAU working group on active B stars (WGABS).

### **XShootU WG12 chair**

Dec 2020 – date

Point of contact (chair) of the pulsations WG12 of ULLYSES targets within the [XShootU](#) collaboration.

### **Co-PI of MOBSTER collaboration**

Nov 2020 – date

Together with PI A. David-Uraz and co-PI C. Neiner, responsible for maximising the scientific productivity of the [MOBSTER](#) collaboration, which leverages TESS data to study massive magnetic stars.

### **SHOC/SAAO and MAIA/Mercator pipeline developer**

Jul 2020 – date

Principal author of the [TEA-PHOT](#) pipeline for the [SHOC/SAAO](#) and [MAIA/Mercator](#) instruments, which is published (Bowman & Holdsworth, 2019, A&A, 629, A21) and fully endorsed as the go-to reduction pipeline.

### **BEST member**

Sep 2019 – date

Member (non-voting) of the [BRITE Executive Science Team](#) (BEST) for the BRITE-constellation space mission.

### **CubeSpec Space Mission**

Jan 2019 – date

Scientific advisor and consortium member for the massive star asteroseismology science case of the [CubeSpec](#) cubesat mission being built by KU Leuven in collaboration with ESA and private contractors.

## **Peer-Review and Grant Panel Experience:**

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### **Grant Funding Reviewer**

Sep 2018 – date

Invited 10+ times to review small and large research grant funding applications for national and international research councils, for example STFC in the UK.

### **Journal Peer Reviewer**

Mar 2016 – date

Invited 50+ times as peer reviewer of publications in international journals: Nature Astronomy, Nature Communications, A&A, MNRAS, ApJ, AJ, PASP, PASA, FRAS, OJA, Galaxies, and JAAVSO.

### **ESO OPC expert panel member**

Sep 2021 to Jan 2023

Observing Program Committee (OPC) expert panel member in panel D (Stellar Evolution) for European Southern Observatory (ESO) proposal semesters P109 – P111. Co-Chair of panel in semester P111.

### **PLATO external reviewer**

Nov 2020

External reviewer for the on-ground data processing algorithms on behalf of the complementary science program (WP16) of the ESA PLATO mission.

## Academic Journal Editor Experience

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### Review Editor, MDPI Galaxies

Sep 2023 – date

Review editor and [editorial board member](#) for the MDPI journal *Galaxies*.

### Associate Editor, Frontiers

Feb 2021 – date

Editorial board member and [Associate Editor](#) for the journal *Frontiers in Astronomy and Space Sciences*.

### Special Issue Editor, MDPI Galaxies

Feb 2022 – Oct 2023

Guest editor for the [Special Issue](#) on Stellar Structure and Evolution for the MDPI journal *Galaxies*.

## Previous Responsibilities:

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### Good Vibrations seminar series

Nov 2020 – July 2026

Steering committee member for the [Good Vibrations](#) seminar series, which provides opportunities for PhD students to share their research internationally.

### WG Chair, Arago space mission

Jan 2022 – Oct 2025

Chair of the 'Hot (BA) Stars' working group for the [Arago mission](#), which was a candidate M7 ESA space mission on UV+Visible spectropolarimetry of stars across the HR diagram. Successful phase 1 application for M7 call, but unsuccessful at phase 2 in Nov 2022. Unsuccessful phase 1 application in ESA's M8 call.

### RAS ECN committee member

Jun 2020 – Mar 2023

Founding member and secretary of the [Early Career Network \(ECN\)](#) of the Royal Astronomical Society (RAS).

## VIII. Professional and Learned Societies

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### **Fellow of the Higher Education Academy**

Jun 2024 – date

Elected a fellow of the [Higher Education Academy](#) (FHEA), which permits the use of the post-nominal FHEA.

### **International Astronomical Union**

Jan 2020 – date

Elected an individual member of the [International Astronomical Union](#) (IAU).

### **European Astronomical Society**

Apr 2019 – date

Member of the European Astronomical Society (EAS).

### **Royal Astronomical Society**

Oct 2013 – date

Elected a fellow of the Royal Astronomical Society (RAS), which permits the use of the post-nominal FRAS.

### **Institute of Physics**

Oct 2013 – date

Member of the Institute of Physics (IOP), which permits the use of the post-nominal MInstP.

## IX. Observing Projects and Experience

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Successful telescope proposals as (co-)PI with competitive ground-based observatories totalling **2200+ hours** and targeting 2300+ massive stars, which include ESO's ESPRESSO/VLT, ESA's PLATO, and NASA's TESS mission. A total of **120+** nights of first-hand experience at world-class telescopes.

### European Southern Observatory (ESO)

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*La Silla Paranal Observatory, Chile*

→ 14 nights of observing experience as visiting astronomer in December 2019 using the FEROS spectrograph mounted on the 2.2-m MPG/ESO telescope at La Silla observatory.

→ 4 nights of observing experience as visiting astronomer in June 2024 using the ESPRESSO spectrograph on UT1 of the VLT at Paranal observatory.

- Co-I of programme obtaining spectropolarimetry of pulsating massive stars with HARPSpol (*117.2ANP*; 48 hr; PI Thomson-Paressant).
- Co-I of programme obtaining multi-epoch UVES spectroscopy of massive contact binaries identified using TESS photometry (*115.286U and 116.28TN*; 52+69 hr; PI Abdul-Masih).
- PI of (visitor mode) programme obtaining time-series spectroscopy of the pulsating massive star zeta Oph with ESPRESSO (*113.26B9.001*; 40 hr; PI Bowman).
- Co-I of large programme obtaining multi-epoch spectroscopy of massive stars in the SMC with FLAMES for the BLOeM consortium (*112.25R7*; 120 hr; PI Shenar).
- Co-I of two programmes obtaining Gravity interferometry of the eclipsing Be binary system HD 93683 (*109.23H0 and 110.246X*; 14 hr; PI Bodensteiner).
- Co-PI of programme obtaining multi-epoch high-resolution spectroscopy of gamma Doradus stars with UVES (*106.21S8*; 21 hr; PI Christophe).
- Co-I of programme obtaining spectropolarimetry of magnetic candidate O-type stars with HARPS (*106.21JB*; 8 hr; PI Barron).
- Co-PI of (visitor mode) programme obtaining multi-epoch high-resolution spectroscopy of massive binary stars with FEROS (*0106.A-9106*; 90 hr; PI Aerts).
- Co-PI of large programme obtaining multi-epoch high-resolution spectroscopy of massive stars with UVES (*1104.D-0230*; 120 hr; PI Tkachenko).
- Co-PI of (visitor mode) programme obtaining multi-epoch high-resolution spectroscopy of massive stars with FEROS (*0104.A-9001*; 120 hr; PI Aerts).
- Co-I of DDT programme obtaining phase-resolved high-resolution spectroscopy of the pulsating eclipsing binary system U Gru with UVES (*103.200F*; 4 hr; PI Johnston).

### PLATO space mission

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*ESA*

- PI of proposal for imagerettes of 121 LMC massive stars to study low-metallicity SLF variability (ID52)
- Co-PI of proposal to study stochastic low-frequency variability in 600+ massive stars (ID21; PI Van Daele)
- Co-PI of proposal of asteroseismology of binary interaction products (ID39; PI Henneco)
- Co-PI of forward asteroseismic modelling of pulsating massive stars proposal (ID35; PI Scott)
- Co-I of proposal on instability strips of magnetic pulsating early-type stars (ID67; PI Thomson-Paressant)
- Co-I of proposal on interior magnetic fields of pulsating early-type stars (ID121; PI Neiner)
- Co-I of proposal to create a census of detached eclipsing binaries (ID131; PI Southworth)

## Transiting Exoplanet Survey Satellite (TESS)

NASA

- PI of six TESS Guest Investigator proposals obtaining high-precision and short-cadence time series photometry of massive stars: cycle 3 in 2020 (*GO3059; 1058 stars; PI Bowman*), cycle 4 in 2021 (*GO4074; 1618 stars; PI Bowman*), cycle 5 in 2022 (*GO5036; 1818 stars; PI Bowman*), cycle 6 in 2023 (*GO6037; 1594 stars; PI Bowman*), cycle 7 in 2024 (*GO7037; 2314 stars; PI Bowman*), and cycle 9 in 2026 (*GO9150; 2027 stars; PI Bowman*).
- Co-I of multiple TESS Guest Investigator proposals obtaining high-precision and short-cadence time series photometry of intermediate- and high-mass stars in cycles 1–4.

## Canada France Hawaii Telescope (CFHT)

Hawai'i, USA

- Co-I of ESPaDOnS programme in semester 26B for magneto-asteroseismology of  $\beta$  Cep, SPB, and  $\delta$  Sct stars (K1-05-00057; 28 hr; PI Thomson-Paressant)
- Co-I of ESPaDOnS programme in semester 26A for magneto-asteroseismology of  $\beta$  Cep, SPB, and  $\delta$  Sct stars (K1-04-00137; 17 hr; PI Thomson-Paressant)
- Co-I of ESPaDOnS programme in semester 25B for Zeeman-Doppler imaging and magneto-asteroseismology of k02 Pup and HD 49198 (K1-04-00011; 20 hr; PI Gutteridge)
- Co-I of ESPaDOnS programme in semester 25B for magneto-asteroseismology of TESS pulsating variable stars (K1-04-00010; 21 hr; PI Thomson-Paressant)
- Co-I of ESPaDOnS programme in semester 25A for spectropolarimetry of the magnetic pulsator HD 183339 (K1-03-00170; 11 hr; PI Gutteridge)
- Co-I of ESPaDOnS programme in semester 24B for spectropolarimetry of early-type pulsating stars (K1-03-00094; 24 hr; PI Gutteridge)
- Co-I of ESPaDOnS programme in semester 23A for spectropolarimetry of the magnetar progenitor HD 45166 (K1-01-00061; 24 hr; PI Wade)
- Co-I of ESPaDOnS programme in semester 23A for spectropolarimetry of early-type pulsating stars (K1-01-00040; 22 hr; PI Neiner)

## Mercator Observatory

La Palma, Spain

→ 65 nights of observing experience using the HERMES and MAIA instruments on the 1.2-m Mercator telescope between 2017 and 2022, which included visitor mode and service observing.

- PI of HERMES large program awarded 1000+ hr across 2022–2025 to obtain multi-epoch spectroscopy of pulsating massive stars discovered by TESS.
- PI of HERMES (visitor mode) proposal awarded 80 hr in semester 2021a to obtain time-series spectroscopy of high-mass pulsating eclipsing binaries discovered by TESS.
- PI of MAIA (visitor mode) proposal awarded 120 hr in semester 2020b to obtain short-cadence multi-colour time-series photometry of roAp stars observed by K2 and TESS.
- PI of HERMES proposal awarded 70 hr in semester 2018b, 35 hr in semester 2019a and 90 hr in semester 2019b to obtain spectroscopy of Ap stars being observed by TESS.
- PI of a HERMES proposal awarded 60 hr in semester 2018a to obtain accurate stellar parameters for pulsating B, A and F stars in the *Kepler* field for forward seismic modelling.
- PI of a HERMES proposal awarded 40 hr in semester 2017a and 20 hr in semester 2018a to study high-mass companions to  $\delta$  Sct stars in binary systems discovered using pulsation timing.

## **Stellar Oscillation Network Group (SONG)**

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*Tenerife, Spain & Australia*

- PI of programme in period 13 to obtain time-series spectroscopy of high-mass pulsating eclipsing binaries (23.AST-07; 260 hr; PI Bowman).

## **South African Astronomical Observatory (SAAO)**

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*Sutherland, South Africa*

→ 21 nights of observing experience as visiting astronomer in May and June 2017 using the SHOC imager mounted on the 1-m telescope at SAAO to obtain high-cadence photometry of roAp stars.

- PI of proposal to obtain high-cadence SHOC photometry of candidate roAp stars in May 2018.

## **Moses Holden Telescope (MHT)**

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*Alston Observatory, Preston, UK*

→ 20 nights of observing experience using the imager on the 0.75-m MHT in 2016 and 2017, and involved in installation, first-light measurements, and co-wrote the first version of the user operational “cookbook”.

## **William Herschel Telescope (WHT)**

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*La Palma, Spain*

- PI of service time proposal in 2016 to gain accurate atmospheric parameters for 23  $\delta$  Sct stars observed by the *Kepler* mission.

## X. Teaching and Supervision Experience

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Supervisor of **4** ongoing PhD students, co-supervisor of **2** ongoing PhD students. Successfully supervised **2** PhD students to completion, and member of an additional **10** international PhD progress and/or examination (jury) committees. Supervisor of **3** postdoctoral research associates at Newcastle University. Supervisor of **4** ongoing and **7** masters students to completion, and member of an additional **5** international MSc thesis examination committees.

### Externally Recruited Postdocs

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- Dr. Jan Henneco June 2025 – date
- Dr. Keegan Thomson-Paressant Feb 2025 – date
- Dr. Laura Scott Sep 2024 – date

### PhD Theses at Newcastle University, UK

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- **Supervisor** of Pieterjan Van Daele Oct 2024 – date  
*Stochastic low-frequency variability in massive stars*
- Progress committee member of Niek Wielders Oct 2024 – date  
Supervisor: Dr. James Nightengale  
*Probing galaxies' hidden depths with gravitational lensing*
- **Supervisor** of Logan Dennis Sep 2024 – date  
*Forward modelling of massive pulsating eclipsing binaries*
- **Co-Supervisor** of Betsy Parnham Sep 2024 – date  
*Dynamical processes in stellar interiors*
- **Co-Supervisor** of Lucas Corrigan Sep 2024 – date  
*MHD simulations of IGWs in massive stars*
- **Supervisor** of Ankur Kalita Apr 2024 – date  
*Spectroscopic and asteroseismic modelling of line profile variability in massive stars*
- **Supervisor** of Federica Nardini Jan 2024 – date  
*Asteroseismology of massive binary systems*

### PhD Theses at KU Leuven, Belgium

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- Progress and examination committee (jury) member of Joris Hermans Sep 2019 – Nov 2023  
Supervisor: Prof. Rony Keppens  
*Understanding the influence of cooling curves and flow on thermal instability*
- **Co-Supervisor** of Jordan Van Beeck Sep 2019 – Sep 2023  
*Asteroseismology of Kepler B stars: internal magnetism and nonlinear mode coupling*
- **Supervisor** of Siemen Burssens Sep 2018 – July 2022  
*Massive star asteroseismology with K2 and TESS*
- Progress and examination committee (jury) member of Joey S. G. Mombarg Feb 2018 – Feb 2022  
Supervisor: Prof. Conny Aerts  
*Asteroseismic modelling of intermediate-mass stars*
- Long-stay host research supervisor of Mariel Lares-Martiz Sep 2019 – Dec 2019  
*Non-linear terms in delta Scuti stars power spectra*

## PhD theses as external committee member

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- Progress committee member of Lucas Barrault Oct 2024 – date  
Supervisor: Prof. Lisa Bugnet, ISTA, Vienna, Austria  
*Unveiling the structure and dynamics of the deep convective core–radiative zone boundary throughout stellar evolution*

## Independent Examiner of PhD theses

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- External PhD thesis examiner (referee) of Alejandro Ramón Ballesta May 2025  
Supervisor: Dr. Javier Pascual, Spanish National Research Council (CSIC), Spain  
*Wavelet analysis applied to PLATO light curves*
- Internal PhD thesis examiner (jury) of Patrick O'Neill Jan 2025  
Supervisor: Dr. Adam Ingram, Newcastle University, UK  
*Probing compact objects through fast timing techniques*
- External PhD thesis examiner (referee) of Abel de Burgos Oct 2024  
Supervisor: Dr. Sergio Simón-Díaz, IAC, Tenerife, Spain  
*On the evolutionary nature of massive B-type supergiants: a modern empirical reappraisal using data from IACOB, Gaia and TESS*
- External PhD thesis examiner (jury) of Keegan Thomson-Paressant Sep 2024  
Supervisor: Dr. Coralie Neiner, Paris Observatory, France  
*Magnetism in  $\delta$  Scuti stars*
- Internal PhD thesis examiner (jury) of Mathias Michielsen Nov 2022  
Supervisor: Prof. Conny Aerts, KU Leuven, Belgium  
*Forward seismic modelling of B-type stars*
- Internal PhD thesis examiner (jury) of Camilla Scolini May 2020  
Supervisor: Prof. Stefaan Poedts, KU Leuven, Belgium  
*Magnetised coronal mass ejections: evolution from the Sun to 1 AU and geo-effectiveness*

## Master students at Newcastle University, UK

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- **Supervisor** of Joe Asal Sep 2025 – June 2026  
Co-supervisors: Keegan Thomson-Paressant and Laura Scott  
*Asteroseismology of  $\beta$  Cep stars with TESS*

## MSc Theses at KU Leuven, Belgium

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- **Supervisor** of Pieterjan Van Daele Sep 2022 – Jun 2023  
*New algorithms to extract blended TESS photometry of massive stars*
- Examination committee member (reader) of Thijs Verhaeghe Jun 2022  
*A target scheduling heuristic for CubeSpec*
- **Supervisor** of Stijn Rutten Sep 2021 – Sep 2022  
*A new user-friendly aperture photometry pipeline for MAIA: variability in pulsating stars*
- **Supervisor** of Nagaraj Vernekar Sep 2020 – Sep 2021  
*On the photometric and spectroscopic variability of Be stars: the case of HD 93683*
- Examination committee member (reader) of Anne Daniels Jun 2021  
*Permutation entropy and statistical complexity to characterise space plasmas*

- Examination committee member (reader) of Mariya Nizovkina Jun 2021  
*Investigating the effect of microturbulent velocity on mass discrepancy in the binary system V380 Cyg*
- Examination committee member (reader) of Tinatin Baratashvili Jun 2020  
*On the effect of grid stretching and AMR on inner heliospheric solar wind and CME evolution simulations*
- **Supervisor** of Joris Hermans Sep 2018 – Jun 2019  
*Testing stellar evolution with selected high-amplitude delta Scuti stars*
- **Supervisor** of Sven Nys Sep 2018 – Jun 2019  
*Asteroseismic modelling of gravity modes in selected intermediate-mass stars*
- **Co-Supervisor** of Jordan Van Beeck Sep 2018 – Jun 2019  
*The influence of an interior magnetic field on gravity-mode oscillations of intermediate-mass stars*
- Examination committee member (reader) of Mathias Michielsen Jun 2018  
*Comparing oscillation frequencies of stars with a convective core: Impact of varying input physics*

### Postgraduate Teaching: MSc modules

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- **Module Leader and Lecturer**, *KU Leuven, Belgium* Sep 2022 – Aug 2023  
Responsible person for delivering MSc Asteroseismology course (30 students; 6 ECTS).
- **Module Leader and Lecturer**, *KU Leuven, Belgium* Sep 2019 – Aug 2023  
Responsible person for delivering the annual MSc thesis defence preparation course (20+ students).

### Undergraduate Teaching: BSc and MPhys modules

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- **Module Leader and Lecturer**, *Newcastle University, UK* Sep 2025 – date  
Third-year undergraduate stellar structure and evolution module (PHY3040; 30+ students; 10 credits).
- **Bachelor and master student projects**, *Newcastle University, UK* Sep 2023 – Sep 2025  
Supervision of multiple bachelor and master student (group) projects.
- **Guest Lecturer**, *University of York, UK* Apr 2024  
Invited lecture on massive star evolution and asteroseismology for MPhys course of Dr. Emily Brunsden.
- **Bachelor and master student projects**, *KU Leuven, Belgium* Sep 2017 – Aug 2023  
Supervision of multiple bachelor and master student (group) projects.
- **Guest Lecturer**, *University of Innsbruck, Austria* May 2021  
Invited lecture on massive stars and asteroseismology for MSc course of Prof. Konstanze Zwintz.
- **Module Examiner**, *KU Leuven, Belgium* Sep 2017 – Aug 2019  
Examiner for the Bachelor science communication course.
- **Module Leader and Lecturer**, *University of Lancashire, UK* Sep 2016 – Jan 2017  
Module leader and lecturer for first-year undergraduate 'Introduction to Statistics' (MA1861; 30 students; 10 credits), 'Stellar Structure and Evolution' (AA1051; 25 students; 10 credits), and second-year astronomy laboratories at UCLan's Alston observatory (AP2060; 25 students; 10 credits).

## XI. Scientific Conferences and Workshops

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Attendance of **65** international conferences and workshops.

|                                                                                        |                       |
|----------------------------------------------------------------------------------------|-----------------------|
| <b>NAM2026</b> , <i>Birmingham University, UK</i>                                      | 20–24 July 2026       |
| <b>TASC10/KASC17 workshop</b> , <i>Aarhus University, Denmark</i>                      | 6–10 July 2026        |
| <b>PLATO Complementary Science Workshop</b> , <i>KU Leuven, Belgium</i>                | 5–6 Feb 2026          |
| <b>PLATO Week 17</b> , <i>KU Leuven, Belgium</i>                                       | 2–4 Feb 2026          |
| <b>HITS Winter workshop</b> , <i>HITS, Heidelberg, Germany</i>                         | 15–17 Dec 2025        |
| <b>HAYDN-1 workshop</b> , <i>(Virtual), University of Bologna, Italy</i>               | 14–16 Jul 2025        |
| <b>TASC9/KASC16 workshop</b> , <i>ISTA, Vienna, Austria</i>                            | 7–11 Jul 2025         |
| <b>Binaries conference</b> , <i>Keele University, UK</i>                               | 30 June – 4 July 2025 |
| <b>BRIDGCE-IRENA meeting</b> , <i>York University, UK</i>                              | 4–6 June 2025         |
| <b>Stellar Hydro Days VI</b> , <i>University of Victoria, Canada</i>                   | 12–16 May 2025        |
| <b>VVS conference</b> , <i>Blankenberge, Belgium</i>                                   | 5–6 Oct 2024          |
| <b>LENAH workshop</b> , <i>KU Leuven, Belgium</i>                                      | 11–13 Sep 2024        |
| <b>Nordita workshop</b> , <i>Stockholm, Sweden</i>                                     | 26–30 Aug 2024        |
| <b>BRITE conference</b> , <i>Vienna University, August</i>                             | 20–23 Aug 2024        |
| <b>TASC8/KASC15 workshop</b> , <i>Porto University, Portugal</i>                       | 15–19 Jul 2024        |
| <b>Owocki-Fest</b> , <i>KU Leuven, Belgium</i>                                         | 8–12 Jul 2024         |
| <b>XShootU workshop</b> , <i>KU Leuven, Belgium</i>                                    | 3–5 Jul 2024          |
| <b>B-fields conference</b> , <i>NAJO, Tokyo, Japan</i>                                 | 25–29 Mar 2024        |
| <b>BLOeM workshop</b> , <i>(Virtual), KU Leuven, Belgium</i>                           | 5–6 Mar 2024          |
| <b>LENAH workshop</b> , <i>Newcastle, UK</i>                                           | 12–14 Dec 2023        |
| <b>BRIDGCE/IReNA consortium meeting</b> , <i>Edinburgh, UK</i>                         | 11–13 Sep 2023        |
| <b>TASC7/KASC14</b> , <i>Honolulu, Hawai'i, USA</i>                                    | 17–21 Jul 2023        |
| <b>National Astronomy Meeting (NAM)</b> , <i>Cardiff University, UK</i>                | 3–7 Jul 2023          |
| <b>Lorentz Workshop</b> , <i>Leiden University, the Netherlands</i>                    | 26–30 Jun 2023        |
| <b>SDSS-V/IReNA/CeNAM Science Festival</b> , <i>KU Leuven, Belgium</i>                 | 3–7 Apr 2023          |
| <b>VFTS</b> , <i>Garching, Germany</i>                                                 | 27–29 Mar 2023        |
| <b>TASC6/KASC13</b> , <i>Leuven, Belgium</i>                                           | 11–15 Jul 2022        |
| <b>IAUS361: Massive Stars Near and Far</b> , <i>Ballyconnell, Ireland</i>              | 8–13 May 2022         |
| <b>KITP program</b> , <i>UCSB, California, USA</i>                                     | 11 Oct – 17 Dec 2021  |
| <b>TESS SciCon II</b> , <i>(Virtual), MIT, USA</i>                                     | 2–6 Aug 2021          |
| <b>BRITE-related Science Meeting</b> , <i>(Virtual), Innsbruck University, Austria</i> | 12 Jul 2021           |
| <b>EAS 2021</b> , <i>(Virtual), Leiden University, the Netherlands</i>                 | 28 Jun – 2 Jul 2021   |
| <b>IAUS361: symposium on massive stars</b> , <i>(Virtual), DIAS, Ireland</i>           | 3–7 May 2021          |

|                                                                                               |                     |
|-----------------------------------------------------------------------------------------------|---------------------|
| <b>OBA stars: Variability and Magnetic Fields</b> , ( <i>Virtual</i> ), <i>St. Petersburg</i> | 26–30 Apr 2021      |
| <b>Pulsations in Multiple Systems</b> , ( <i>Virtual</i> ), <i>University of Surrey, UK</i>   | 18–22 Jan 2021      |
| <b>MOBSTER-1</b> , ( <i>Virtual</i> ), <i>University of Delaware, USA</i>                     | 13–17 Jul 2020      |
| <b>EAS 2020</b> , ( <i>Virtual</i> ), <i>Leiden University, the Netherlands</i>               | 29 Jun – 3 Jul 2020 |
| <b>Stars and their Variability</b> , <i>Vienna University, Austria</i>                        | 19–23 Aug 2019      |
| <b>TESS Sci Con I</b> , <i>MIT, Cambridge, USA</i>                                            | 29 Jul – 2 Aug 2019 |
| <b>TASC5/KASC12</b> , <i>MIT, Cambridge, USA</i>                                              | 22–26 Jul 2019      |
| <b>Stellar Hydro Days V</b> , <i>Exeter University, UK</i>                                    | 24–28 Jun 2019      |
| <b>STFC/MAMSIE workshop</b> , <i>KU Leuven, Belgium</i>                                       | 2–4 Apr 2019        |
| <b>Kepler/K2 Sci Con V</b> , <i>Glendale, California, USA</i>                                 | 4–8 Mar 2019        |
| <b>TESS data workshop</b> , <i>KU Leuven, Belgium</i>                                         | 5–9 Nov 2018        |
| <b>STFC/MAMSIE workshop</b> , <i>KU Leuven, Belgium</i>                                       | 29–31 Oct 2018      |
| <b>MASSIVE star meeting</b> , <i>KU Leuven, Belgium</i>                                       | 4–6 Oct 2018        |
| <b>PHOST</b> , <i>Banyuls-sur-mer, France</i>                                                 | 3–7 Sep 2018        |
| <b>TASC4/KASC11</b> , <i>Aarhus University, Denmark</i>                                       | 8–13 Jul 2018       |
| <b>Statistics workshop</b> , <i>KU Leuven, Belgium</i>                                        | 11 Jun 2018         |
| <b>STFC/MAMSIE workshop</b> , <i>Newcastle University, UK</i>                                 | 5–8 Jun 2018        |
| <b>Belgian contact group meeting</b> , <i>Brussels, Belgium</i>                               | 4 Jun 2018          |
| <b>MAMSIE/STFC workshop</b> , <i>KU Leuven, Belgium</i>                                       | 14–16 Mar 2018      |
| <b>TESS data workshop</b> , <i>KU Leuven, Belgium</i>                                         | 6–8 Dec 2017        |
| <b>MAMSIE/STFC workshop</b> , <i>KU Leuven, Belgium</i>                                       | 12–15 Sep 2017      |
| <b>MESA Summer school</b> , <i>UCSB, California, USA</i>                                      | 14–18 Aug 2017      |
| <b>TASC3/KASC10</b> , <i>Birmingham University, UK</i>                                        | 17–21 Jul 2017      |
| <b>STARS2016</b> , <i>Windermere, UK</i>                                                      | 11–15 Sep 2016      |
| <b>TASC2/KASC9 workshop</b> , <i>Terceira-Açores, Portugal</i>                                | 27 Jun – 1 Jul 2016 |
| <b>National Astronomy Meeting (NAM)</b> , <i>Nottingham University, UK</i>                    | 11–15 Jul 2016      |
| <b>STFC spectroscopy school</b> , <i>Queen's University Belfast, UK</i>                       | 31 Aug – 4 Sep 2015 |
| <b>KASC8/TASC1 workshop</b> , <i>Aarhus University, Denmark</i>                               | 15–19 Jun 2015      |
| <b>RAS specialist discussion meeting</b> , <i>RAS, London, UK</i>                             | 8 May 2015          |
| <b>K2 data workshop</b> , ( <i>Virtual</i> ) <i>Aarhus University, Denmark</i>                | 10–11 Nov 2014      |
| <b>Ecole Evry Schatzman 2014</b> , <i>Roscoff, France</i>                                     | 28 Sep – 3 Oct 2014 |
| <b>CoRoT3/KASC7 meeting</b> , <i>Toulouse, France</i>                                         | 6–11 Jul 2014       |
| <b>Spectroscopy workshop</b> , <i>Aarhus University, Denmark</i>                              | 19–23 May 2014      |

## XII. Conference Talks, Seminars and Colloquia

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Total of **16 invited** and **20 contributed** talks at international conferences, and **32** seminars/colloquia.

### Conference Talks

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- NAM2026 (Contributed), *University of Birmingham, UK* 20 July 2026
- HITS Winter workshop (Contributed), *HITS, Heidelberg, Germany* 15 Dec 2025
- HAYDN-1 workshop (**Invited**), *University of Bologna, Italy* 14 Jul 2025
- Binaries conference (**Invited**), *Keele University, UK* 3 Jul 2025
- BRIDGCE-IReNA meeting (**Invited**), *York University, UK* 6 June 2025
- Stellar Hydro Days VI (**Two Invited**), *Victoria, Canada* 14 and 16 May 2025
- VVS Conference (**Invited**), *Blankenberge, Belgium* 5 Oct 2024
- Nordita workshop (**Invited**), *Stockholm, Sweden* 27 Aug 2024
- BRITE conference workshop (Two Contributed), *Vienna, Austria* 22 Aug 2024
- TASC8/KASC15 conference (Contributed), *Porto, Portugal* 19 Jul 2024
- B-fields 2024 conference (Contributed), *Tokyo, Japan* 28 Mar 2024
- George Darwin Lecture (**Invited**), *RAS, London, UK* 12 Jan 2024
- BRIDGCE/IReNA meeting (Contributed), *Edinburgh, UK* 11 Sep 2023
- TASC7/KASC14 (Contributed), *Honolulu, Hawai'i, USA* 18 Jul 2023
- National Astronomy Meeting (NAM) (Contributed), *Cardiff, UK* 5 Jul 2023
- IAUS361: Massive Stars Near and Far (Contributed), *Ballyconnell, Ireland* 10 May 2022
- Probes of Transport in Stars (**Invited**), *KITP, UCSB, USA* 15 Nov 2021
- Probes of Transport in Stars (**Invited**), *KITP, UCSB, USA* 12 Oct 2021
- TESS SciCon II (Contributed), *Virtual (hosted by MIT, USA)* 3 Aug 2021
- BRITE-related Science Meeting (**Invited**), *Virtual (hosted by Uni. Innsbruck, Austria)* 12 Jul 2021
- IAUS361 symposium (Contributed), *Virtual (hosted by DIAS, Ireland)* 3 May 2021
- OBA stars: variability and magnetic fields (**Invited**), *Virtual (hosted by St. Petersburg)* 30 Apr 2021
- PIMMS workshop (**Invited**), *Virtual (hosted by University of Surrey, UK)* 18 Jan 2021
- Stars and their Variability (**Invited**), *Vienna, Austria* 19 Aug 2019
- TESS Sci Con I (Contributed), *MIT, Cambridge, USA* 30 Jul 2019
- TASC5/KASC12 (Contributed), *MIT, Cambridge, USA* 23 Jul 2019
- Stellar Hydro Days V (Contributed), *Exeter, UK* 26 Jun 2019
- Kepler/K2 Sci Con V (Contributed), *Glendale, California, USA* 7 Mar 2019
- MASSIVE star meeting (Contributed), *Leuven, Belgium* 4 Oct 2018
- PHOST conference (Contributed), *Banyuls-sur-mer, France* 6 Sep 2018
- TASC4/KASC11 workshop (**Invited**), *SAC, Aarhus University, Denmark* 13 Jul 2018
- STARS2016 conference (Contributed), *Windermere, UK* 14 Sep 2016

- KASC8/TASC1 workshop (Contributed), *SAC, Aarhus University, Denmark* 15 Jun 2015
- RAS specialist discussion meeting (**Invited**), *RAS, London, UK* 8 May 2015

## Seminars and Colloquia

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- Sheffield University, UK 29 April 2026
- IAU Astrostats working group (virtual) [[recording](#)] 10 March 2026
- Armagh Observatory, UK 29 May 2025
- Northumbria University, UK 5 Feb 2025
- IRENA, USA (virtual) [[recording](#)] 31 Jan 2025
- Newcastle University, UK 22 Nov 2024
- Chalmers University of Technology, Sweden 6 Nov 2024
- Innsbruck University, Austria 12 Mar 2024
- HITS, Heidelberg, Germany 22 Nov 2023
- Newcastle University, UK 1 Nov 2023
- Newcastle University, UK 25 Oct 2023
- Amsterdam University, the Netherlands 10 May 2023
- European Southern Observatory (ESO), Santiago, Chile 9 Feb 2023
- Thüringer Landessternwarte (TLS) Tautenburg, Germany (virtual) 24 Nov 2022
- Newcastle University, UK 2 Nov 2022
- Chinese University of Hong Kong (virtual) 2 Jun 2022
- MPA, Germany (virtual) 20 Apr 2022
- University of Geneva, Switzerland (virtual) 14 Apr 2022
- Sheffield University, UK (virtual) 6 Oct 2021
- KU Leuven, Belgium [[recording](#)] 1 Oct 2021
- Keele University, UK (virtual) 19 May 2021
- Nicolaus Copernicus Astronomical Center, Poland (virtual) 21 Apr 2021
- KITP, California, USA (virtual) 16 Dec 2020
- KU Leuven, Belgium 22 Mar 2019
- Newcastle University, UK 6 Jun 2018
- Université Libre de Bruxelles, Belgium 19 Apr 2018
- KU Leuven, Belgium 2 Mar 2018
- Royal Observatory of Belgium, Belgium 16 Nov 2017
- University of Lancashire, UK 15 Jun 2016
- SAC, Aarhus University, Denmark 2 May 2016
- University of Lancashire, UK 15 Jul 2015
- Keele University, UK 4 Sep 2014

## XIII. Public Engagement and Outreach

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Extensive experience in public engagement and outreach in science and astronomy, having organised **100+** outreach events for schools, amateur astronomer societies and the general public, as well as virtual activities around the world. Demonstrable positive impact on thousands of people in terms of participant satisfaction and engagement with astronomy at various levels. Dedicated to continuing to provide high-calibre outreach activities throughout career. Examples of outreach and engagement activities to date include:

### Newcastle University (2023 – date)

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- Contributor to the [Audio Universe](#) astro-sonification project led by Newcastle and Portsmouth universities.
- Speaker and exhibitor at the [Moon Palace](#) visit to Newcastle University on 21 Sep 2025 for 100+ attendees.
- Stars and Planets exhibition and talk at [Great North Night](#) on 6 Dec 2024 for 200+ attendees.
- Invited public lecture for amateur observing society at Kielder castle on 2 Nov 2024 for 80+ attendees.
- Invited public lecture for the [Flemish Astronomical Society](#) on 5 Oct 2024 for 200+ attendees.
- Invited [public lecture](#) on asteroseismology at the University of York on 11 April 2024, with 80+ attendees.
- Astronomy engagement (virtual) talk for 30+ undergraduate physics students in Kenya on 7 Feb 2024.
- Astronomy engagement talk for 300+ high-school students aged 14-16 on 31 Jan 2024.
- Participant of the [Skype a Scientist](#) program, with over a dozen online astronomy discussions with international participants, including school classrooms and families around the world.

### KU Leuven, Belgium (2017 – 2023)

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- Teaching Fellow of the [World Science Scholar](#) program of the World Science Foundation for the academic year 2022–2023, in which 60+ gifted and talented high school students from around the world were selected to expand their mathematical abilities with inspiring university-level topics led by world-renowned scientists.
- Participant of the [Scientist@School](#) program, and delivered over a dozen astronomy-themed talks and activities for local Belgian schools, with classes up to approximately 30 students aged 14–18.
- A series of short popular-science videos in collaboration with Huawei and Pint of Science Belgium for the '[5-minute science you never knew](#)' playlist of the 'What Makes it Tick?' YouTube channel.
- Interviewed for the Astronomer job profile for the UK [prospects career advice](#) website in 2021.
- Guest lecturer in stellar physics for the [Vereniging Voor Sterrenkunde Zomerschool](#) for 30+ students aged 16–18 in August 2017, and again in 2018 and 2020.
- Co-author of (Dutch) article for the September 2019 issue of the popular astronomy magazine [Heelal](#).
- Invited speaker at two [Pint of Science](#) events in Brussels on 7 and 21 May 2019, each with 150+ attendees.
- Workshops on space exploration and the solar system at the KU Leuven [Kids University](#), for 30 students aged 8–13 on 5 May 2018 and 22 October 2022.
- Workshops on Exoplanets, Habitability and Host Star Variability for the [Ladies@Science](#) 2017 event, hosted at KU Leuven for 40+ female students aged 14–16 on 19 April 2017.

### University of Lancashire (2013 – 2017)

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Organised dozens of public visits to the 0.7-m telescope at [Alston Observatory](#) and gave an interactive tours of the night sky using the modern planetarium. Visited dozens of primary and secondary schools to give talks and run astronomy-themed group activities for classes of about 30 students aged 6–16.

## XIV. Peer-Reviewed Scientific Publications

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As of 14 September 2026, publication citation metrics:

**Google scholar:** 6864 citations and h-index of 48

**NASA ADS:** 5641 citations and h-index of 45

Total of **19** peer-reviewed papers as first-author and **98** as co-author (of which **28** as second or third author), which include **7** publications in *Nature*, *Nature Astronomy*, and *Science*, and **4** invited single/dual author review papers.

Publications led by PhD students under direct supervision are underlined in gold.

### Submitted papers currently under review:

- C. Eze, G. Handler, M. Vanrespaille, **D. M. Bowman**, C. Aerts, (*submitted, A&A*), ‘*The mass discrepancy problem in massive stars: an inferential perspective*’
- T. Sayada, T. Shenar, R. Ovdia, Z. Z. Katabi, J. Bodensteiner, N. Britavskiy, N. Langer, L. Mahy, P. Marchant, H. Sana, F. R. N. Schneider, S. Vilaza-Dallago, J. I. Villaseñor, **D. M. Bowman**, A. de Burgos Sierra, K. Deshmukh, M. Fabry, Y. Götzberg, M. Gull, G. Holgado, R. G. Izzard, J. Maíz Apellániz, T. Mazeh, L. M. Oskinova, A. Parkosidis, L. R. Patrick, D. Pauli, D. F. Rocha, A. Schootemeijer, C. Schürmann, K. Sen, S. Shahaf, M. Stoop, J. Th. van Loon, X.-T. Xu, (*submitted, A&A*), ‘*Binarity at LOw Metallicity (BLOeM): Double-lined early B-type spectroscopic binaries in the SMC*’
- A. Kemp, E. Sipkova, R. Ochoa-Armenta, **D. M. Bowman**, V. Vanlaer, D. J. Fritzewski, L. IJspeert, A. Tkachenko, C. Aerts, (*submitted, A&A*), ‘*Populations of tidally perturbed pressure-mode pulsators in eclipsing binaries*’
- A. Kalita, M. Abdul-Masih, **D. M. Bowman**, S. Simón-Díaz, (*submitted, MNRAS*), ‘*A new self-consistent 3D approach to modelling macroturbulence in rotating massive stars with SPAMMS*’
- F. Nardini, **D. M. Bowman**, J. Bodensteiner, P. Van Daele, L. J. A. Scott, K. Thomson-Paressant, J. Henneco, H. Sana, (*submitted, MNRAS*), ‘*Characterisation of B-type stars in four young Galactic open clusters. II. Photometric variability with TESS and identification of pulsating stars*’
- C. I. Eze, G. Handler, M. Vanrespaille, A. Kemp, A. Tkachenko, **D. M. Bowman**, C. Aerts, (*submitted, A&A*), ‘*Ensemble pulsational characteristics of eclipsing binaries containing  $\beta$  Cep stars observed by the TESS mission*’
- M. Kliapets, P. Huijse, J. Audenaert, A. Tkachenko, M. Skarka, P. F. X. Gregory, **D. M. Bowman**, S. J. Murphy, P. Agrawal, J. M. Benkö, H. Brinkman, N. E. Janssen, Y. N. E. Eschen, A. Eto, D. J. Fritzewski, A. Kemp, V. Khalack, G. Li, R. Ochoa-Armenta, I. M. Rolo, B. M. Sanz, N. Scheller, R. S. Stanley, K. Thomson-Paressant, E. Plachy, V. Vanlaer, M. Vanrespaille, J. Vrancken, H. Wang, Y. Xia, G. R. Ricker, C. Aerts, (*submitted, A&A*), ‘*Variability classification of TESS targets in LOPS2, the first long-term pointing field of PLATO. Version 1 of the public variability catalogue*’ [[ADS link](#)]

### Accepted papers currently in press:

- M. Abdul-Masih, C. Hawcroft, T. Lechien, S. Simón-Díaz, H. Sana, K. Deshmukh, J. Vrancken, J. I. Villaseñor, J. Müller-Horn, A. J. Kalita, B. Ludwig, J. Bodensteiner, **D. M. Bowman**, A. Escorza, G. Holgado, J. Puls, A. de Vicente, (*in press, A&A*), ‘*SpecFANN: Spectral Fitting via Artificial Neural Networks. A deep learning based FASTWIND emulator and fitting suite*’ [[ADS link](#)]

### Published articles:

- V. Antoci, J. Labadie-Bartz, M. Świąch, O. Dürfeldt-Pedros, S. J. Murphy, D. W. Kurtz, T. R. Bedding, G. Handler, J. Fuller, R.-M. Ouazzani, H. Kjeldsen, L. Fellay, M. G. Pedersen, E. Niemczura, **D. M. Bowman**, M. Deal, P. Mani, (2026), *A&A*, Volume 712, A220, ‘*Wildly Oscillating Stars. Unexplained dense ridge-like frequency agglomerations in A and F type pulsators*’ [[ADS link](#)]

- K. Thomson-Paressant, **D. M. Bowman**, F. Nardini, L. J. A. Scott, J. Bodensteiner, T. Shenar, L. Mahy, G. Handler, N. Shitrit, I. Arcavi, M. Abdul-Masih, S. Simón-Díaz, P. Van Daele, A. J. Kalita, L. Dennis, J. Henneco, H. Sana, H. Van Winckel, A. Tkachenko, (2026), MNRAS, Volume 551, Issue 2, stag1493, '*The symphony of pulsations and binarity among massive stars using HERMES spectroscopy and TESS photometry*' [\[ADS link\]](#)
- J. Southworth, **D. M. Bowman**, (2026), ARA&A, Volume 64, 223-260, '*Pulsations in Binary Star Systems*' (Invited review) [\[ADS link\]](#)
- P. J. Van Daele, **D. M. Bowman**, R. Ovdia, Z. Katabi, J. Bodensteiner, T. Shenar, N. Langer, J. Henneco, A. Kalita, P. A. Crowther, M. Gull, L. Mahy, L. Patrick, D. Pauli, M. Pawlak, (2026), MNRAS, Volume 550, Issue 2, stag1131, '*Binarity at LOw Metallicity (BLOeM): massive star variability revealed using a novel software tool for point-spread function fitting of TESS images*' [\[ADS link\]](#)
- J. Henneco, **D. M. Bowman**, (2026), MNRAS, Volume 549, Issue 4, stag1132, '*The effect of near-core mixing on rejuvenation and the asteroseismic properties of massive accretors*' [\[ADS link\]](#)
- A. Moharana, J. Southworth, K. Pavlovski, A. Miszuda, R. S. Rathour, K. G. Helminiak, F. Marcadon, **D. M. Bowman**, T. Pawar, A. Tkachenko, (2026), MNRAS, Volume 548, Issue 4, stag772, '*V446 Cephei: a  $\beta$  Cep pulsator in a multiple system*' [\[ADS link\]](#)
- D. J. Lennon, S. R. Berlanas, A. Herrero, N. Britavskiy, P. L. Dufton, N. Langer, H. Jin, A. Schootemeijer, A. Menon, J. Bestenlehner, P. Crowther, J. Vink, J. Bodensteiner, T. Shenar, K. Deshmukh, H. Sana, J. Villaseñor, L. Patrick, P. Najarro, A. de Koter, L. Mahy, **D. M. Bowman**, A. Bobrick, C. J. Evans, M. Gull, G. Holgado, Z. Katabi, J. Kubát, P. Marchant, D. Pauli, M. Pawlak, M. Renzo, D. F. Rocha, A. A. C. Sander, T. Sayada, S. Simón-Díaz, M. Stoop, R. Valli, C. Wang, X.-T. Xu, (2026), A&A, Volume 707, A204, '*Binarity at LOw Metallicity (BLOeM): Projected rotational velocities*' [\[ADS link\]](#)
- P. Neuville, H. Sana, A. Tkachenko, P. Royer, J. Windey, A. Vermant, G. Raskin, J. Pember, **D. M. Bowman**, B. Vandenbussche, (2026), A&A, Volume 705, A93, '*The CubeSpec space mission. II. Observational strategy validation*' [\[ADS link\]](#)
- L. J. A. Scott, **D. M. Bowman**, (2026), MNRAS, Volume 545, Issue 3, staf2174, '*Asteroseismology of SPB stars: a comparison of forward asteroseismic modelling results from Kepler and TESS*' [\[ADS link\]](#)
- **D. M. Bowman**, L. Bugnet, (2026), Encyclopedia of Astrophysics, Volume 2, I. Mandel, (eds.), ISBN 978-0-443-21439-4, Elsevier, pp. 133-153, '*Asteroseismology*' (Invited Review) [\[ADS link\]](#)

#### 2025: 10 co-author publications

- A. J. Kalita, **D. M. Bowman**, M. Abdul-Masih, S. Simón-Díaz, (2025), A&A, Volume 703, A2, '*Large-scale variability in macroturbulence driven by pulsations in the rapidly rotating massive star  $\zeta$  Oph from high-cadence ESPRESSO spectroscopy and TESS photometry*' [\[ADS link\]](#)
- V. Vanlaer, **D. M. Bowman**, S. Burssens, S. B. Das, L. Bugnet, S. Mathis, C. Aerts, (2025), A&A, Volume 701, A5, '*Interior rotation modelling of the  $\beta$  Cep pulsator HD 192575 including multiplet asymmetries*' [\[ADS link\]](#)
- H. Sana, T. Shenar, J. Bodensteiner, N. Britavskiy, N. Langer, D. J. Lennon, L. Mahy, I. Mandel, S. E. de Mink, L. R. Patrick, J. I. Villaseñor, M. Dirickx, M. Abdul-Masih, L. A. Almeida, F. Backs, S. R. Berlanas, M. Bernini-Peron, **D. M. Bowman**, V. A. Bronner, P. A. Crowther, K. Deshmukh, C. J. Evans, M. Fabry, M. Gieles, A. Gilkis, G. González-Torà, G. Gräfener, Y. Götzberg, C. Hawcroft, V. Hénault-Brunet, A. Herrero, G. Holgado, R. G. Izzard, A. de Koter, S. Janssens, C. Johnston, J. Josiek, S. Justham, V. M. Kalari, J. Klencki, J. Kubát, B. Kubátová, R. R. Lefever, J. Th. van Loon, B. Ludwig, J. Mackey, J. Maíz Apellániz, G. Maravelias, P. Marchant, T. Mazeh, A. Menon, M. Moe, F. Najarro, L. M. Oskinova, R. Ovdia, D. Pauli, M. Pawlak, V. Ramachandran, M. Renzo, D. F. Rocha, A. A. C. Sander, F. R. N. Schneider, A. Schootemeijer, E. C. Schösser, C. Schürmann, K. Sen, S. Shahaf, S. Simón-Díaz, L. A. C. van Son, M. Stoop, S. Toonen, F.

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- F. Nardini, J. Bodensteiner, H. Sana, L. Mahy, K. Deshmukh, **D. M. Bowman**, (2025), *MNRAS*, Volume 540, Issue 3, pp. 2009–2030, '*Characterisation of B-type stars in four young Galactic open clusters. I. Stellar content and binary properties*' [\[ADS link\]](#)
- H. Rauer, C. Aerts, J. Cabrera, et al. (total of over 800 co-authors, including **D. M. Bowman**, in the PLATO consortium), *Experimental Astronomy*, Volume 59, Issue 3, id. 26, '*The PLATO Mission*' [\[ADS link\]](#)
- J. I. Villaseñor, H. Sana, L. Mahy, T. Shenar, J. Bodensteiner, N. Britavskiy, D. J. Lennon, M. Moe, L. R. Patrick, M. Pawlak, **D. M. Bowman**, P. A. Crowther, S. E. de Mink, K. Deshmukh, M. Fabry, M. Foesneau, G. Holgado, N. Langer, J. Maíz Apellániz, I. Mandel, L. M. Oskinova, M. Renzo, H.-W. Rix, D. F. Rocha, A. A. C. Sander, F. R. N. Schneider, K. Sen, S. Simón-Díaz, J. Th. van Loon, J. S. Vink, (2025), *A&A*, Volume 698, A41, '*Binarity at LOW Metallicity (BLOeM): Enhanced multiplicity in early B-type dwarfs and giants at  $Z = 0.2Z_{\odot}$* ' [\[ADS link\]](#)
- N. Britavskiy, L. Mahy, D. J. Lennon, L. R. Patrick, H. Sana, J. I. Villaseñor, T. Shenar, J. Bodensteiner, M. Bernini-Peron, S. R. Berlanas, **D. M. Bowman**, P. A. Crowther, S. E. de Mink, C. J. Evans, Y. Götzberg, G. Holgado, C. Johnston, Z. Keszthelyi, J. Klencki, N. Langer, I. Mandel, A. Menon, M. Moe, L. M. Oskinova, D. Pauli, V. Ramachandran, M. Renzo, A. A. C. Sander, F. R. N. Schneider, A. Schootemeijer, K. Sen, S. Simón-Díaz, J. Th. van Loon, J. S. Vink, (2025), *A&A*, Volume 698, A40, '*Binarity at LOW Metallicity (BLOeM): Multiplicity of early B-type supergiants in the SMC*' [\[ADS link\]](#)
- L. R. Patrick, D. J. Lennon, F. Najarro, T. Shenar, J. Bodensteiner, H. Sana, P. A. Crowther, N. Britavskiy, N. Langer, A. Schootemeijer, C. J. Evans, L. Mahy, Y. Götzberg, S. E. de Mink, F. R. N. Schneider, A. J. G. O'Grady, J. I. Villaseñor, M. Bernini-Peron, **D. M. Bowman**, A. de Koter, K. Deshmukh, A. Gilkis, G. González-Torà, V. M. Kalari, Z. Keszthelyi, I. Mandel, A. Menon, M. Moe, L. M. Oskinova, D. Pauli, M. Renzo, A. A. C. Sander, K. Sen, M. Stoop, J. Th. van Loon, S. Toonen, F. Tramper, J. S. Vink, C. Wang, (2025), *A&A*, Volume 698, A39, '*Binarity at LOW Metallicity (BLOeM): The multiplicity properties and evolution of BAF-type supergiants*' [\[ADS link\]](#)
- J. Bodensteiner, T. Shenar, H. Sana, N. Britavskiy, P. A. Crowther, N. Langer, D. J. Lennon, L. Mahy, L. R. Patrick, J. I. Villaseñor, M. Abdul-Masih, **D. M. Bowman**, A. de Koter, S. E. de Mink, K. Deshmukh, M. Fabry, A. Gilkis, Y. Götzberg, G. Holgado, R. G. Izzard, S. Janssens, V. M. Kalari, Z. Keszthelyi, J. Kubát, I. Mandel, G. Maravelias, L. M. Oskinova, D. Pauli, V. Ramachandran, D. F. Rocha, M. Renzo, A. A. C. Sander, F. R. N. Schneider, A. Schootemeijer, K. Sen, M. Stoop, S. Toonen, J. Th. van Loon, R. Valli, J. S. Vink, C. Wang, X.-T. Xu, (2025), *A&A*, Volume 698, A38, '*Binarity at LOW Metallicity (BLOeM): Multiplicity properties of Oe and Be stars*' [\[ADS link\]](#)
- N. Janssen, A. Tkachenko, P. Royer, J. De Ridder, D. Seynaeve, C. Aerts, S. Aigrain, E. Planchy, A. Bodi, M. Uzundag, **D. M. Bowman**, D. J. Fritzewski, L. W. IJspeert, G. Li, M. G. Pedersen, M. Vanrespaille, T. Van Reeth, (2025), *A&A*, Volume 694, A185, '*MOCKA – A PLATO mock asteroseismic catalogue: Simulations for gravity-mode oscillators*' [\[ADS link\]](#)

**2024:** 1 first author and 8 co-author publications

- R. Ratnasingam, P. V. F. Edelmann, **D. M. Bowman**, T. M. Rogers, (2024), *ApJ Letters*, Volume 977, Issue 1, L30, '*On the geometry of the near-core magnetic field in massive stars*' [\[ADS link\]](#)
- **D. M. Bowman**, P. Van Daele, M. Michielsen, T. Van Reeth, (2024), *A&A*, Volume 692, A49, '*Photometric detection of internal gravity waves in upper main-sequence stars. IV. Comparable stochastic low-frequency variability in SMC, LMC and Galactic massive stars*' [\[ADS link\]](#)
- T. Shenar, J. Bodensteiner, H. Sana, P. A. Crowther, D. J. Lennon, M. Abdul-Masih, L. A. Almeida, F. Backs, S. R. Berlanas, M. Bernini-Peron, J. M. Bestenlehner, **D. M. Bowman**, V. A. Bronner, N. Britavskiy, A. de Koter, S. E. de Mink, K. Deshmukh, C. J. Evans, M. Fabry, M. Gieles, A. Gilkis, G. González-Torà, G.

- Gräfener, Y. Götberg, C. Hawcroft, V. Hénault-Brunet, A. Herrero, G. Holgado, S. Janssens, C. Johnston, J. Josiek, S. Justham, V. M. Kalari, Z. Z. Katabi, Z. Keszthelyi, J. Klencki, J. Kubát, B. Kubátová, N. Langer, R. R. Lefever, B. Ludwig, J. Mackey, L. Mahy, J. Maíz Apellániz, I. Mandel, G. Maravelias, P. Marchant, A. Menon, F. Najarro, L. M. Oskinova, A. J. G. O'Grady, R. Ovidia, L. R. Patrick, D. Pauli, M. Pawlak, V. Ramachandran, M. Renzo, D. F. Rocha, A. A. C. Sander, T. Sayada, F. R. N. Schneider, A. Schootemeijer, E. C. Schösser, C. Schürmann, K. Sen, S. Shahaf, S. Simón-Díaz, M. Stoop, S. Toonen, F. Tramper, J. Th. van Loon, R. Valli, L. A. C. van Son, A. Vigna-Gómez, J. I. Villaseñor, J. S. Vink, C. Wang, R. Willcox, (2024), *A&A*, Volume 690, A289 '*Binarity at LOw Metallicity (BLOeM): A spectroscopic VLT monitoring survey of massive stars in the SMC*' [\[ADS link\]](#)
- E. Farrell, G. Buldgen, G. Meynet, P. Eggenberger, M.-A. Dupret, **D. M. Bowman**, (2024), *A&A*, Volume 686, A267, '*A method for non-linear inversion of the stellar structure applied to gravity mode pulsators*' [\[ADS link\]](#)
  - W. R. Thompson, F. Herwig, P. R. Woodward, H. Mao, P. Denissenkov, **D. M. Bowman**, S. Blouin, (2024), *MNRAS*, Volume 531, Issue 1, 1316–1337, '*3D hydrodynamic simulations of massive main-sequence stars. II. Convective excitation and spectra of internal gravity waves*' [\[ADS link\]](#)
  - A. J. Frost, H. Sana, L. Mahy, G. Wade, J. Barron, J.-B. Le Bouquin, A. Mérand, F. R. N. Schneider, T. Shenar, R. H. Barba, **D. M. Bowman**, M. Fabry, A. Farhang, P. Marchant, N. I. Morrell, J. V. Smoker, (2024), *Science*, Volume 384, Issue 6692, 214–217, '*A magnetic massive star has experienced a stellar merger*' [\[ADS link\]](#)
  - A. Tkachenko, K. Pavlovski, N. Serebriakova, **D. M. Bowman**, L. IJspeert, S. Gebruers, J. Southworth, (2024), *A&A*, Volume 683, A252, '*Observational mapping of the mass discrepancy in eclipsing binaries. Selection of the sample and its photometric and spectroscopic properties*' [\[ADS link\]](#)
  - K. Zwintz, A. Pigulski, R. Kuschnig, G. A. Wade, G. Doherty, M. Earl, C. Lovekin, M. Müllner, S. Piché-Perrier, T. Steindl, P. G. Beck, K. Bicz, **D. M. Bowman**, G. Handler, B. Pablo, A. Popowicz, T. Rózański, P. Mikołajczyk, D. Baade, O. Koudelka, A. F. J. Moffat, C. Neiner, P. Orleański, R. Smolec, N. St. Louis, W. W. Weiss, M. Wenger, E. Zocłńska, (2024), *A&A*, Volume 683, A49, '*Catalogue of BRITE-Constellation. I. Fields 1 to 14 (November 2013 – April 2016)*' [\[ADS link\]](#)
  - D. L. Holdsworth, M. S. Cunha, M. Lares-Martiz, D. W. Kurtz, V. Antoci, S. Barceló Forteza, P. De Cat, A. Derekas, C. Kayhan, D. Ozuyar, M. Skarka, D. R. Hey, F. Shi, **D. M. Bowman**, O. Kobzar, A. Ayala Gómez, Zs. Bognár, D. L. Buzasi, M. Ebadi, L. Fox-Machado, A. García Hernández, H. Ghasemi, J. A. Guzik, G. Handler, A. Hasanzadeh, R. Jayaraman, V. Khalack, O. Kochukhov, C. C. Lovekin, P. Mikołajczyk, D. Mkrtichian, S. J. Murphy, E. Niemczura, B. G. Olafsson, J. Pascual-Granado, E. Paunzen, N. Positek, A. Ramón-Ballesta, H. Safari, A. Samadi-Ghadim, B. Smalley, Á. Sódor, I. Stateva, J. C. Suárez, R. Szabó, T. Wu, E. Ziaali, W. Zong, (2024), *MNRAS*, Volume 527, Issue 4, 9548–9580, '*TESS Cycle 2 observations of roAp stars with 2-min cadence data*' [\[ADS link\]](#)

**2023:** 2 first author and 11 co-author publications

- **D. M. Bowman**, (2023), *Astrophysics and Space Science*, Volume 368, Issue 12, 107, '*Making waves in massive star asteroseismology*' (**Invited review**) [\[ADS link\]](#)
- **D. M. Bowman**, J. Van Saders, J. Vink, (2023), *Galaxies*, Volume 11, Issue 5, 94, '*The Structure and Evolution of Stars: Introductory Remarks*' [\[ADS link\]](#)
- T. Shenar, G. Wade, P. Marchant, S. Bagnulo, J. Bodensteiner, **D. M. Bowman**, A. Gilkis, N. Langer, A. Nicholas-Chené, L. Oskinova, T. Van Reeth, H. Sana, N. St-Louis, A. Soares de Oliveira, H. Todt, S. Toonen, (2023), *Science*, Volume 381, Issue 6659, 761–765, '*A massive helium star with a sufficiently strong magnetic field to form a magnetar*' [\[ADS link\]](#)
- N. Serebriakova, A. Tkachenko, S. Gebruers, **D. M. Bowman**, T. Van Reeth, L. Mahy, S. Burssens, L. IJspeert, H. Sana, C. Aerts, (2023), *A&A* Volume 676, A85, '*The ESO UVES/FEROS Large Programs of TESS OB pulsators. I. Global stellar parameters from high-resolution spectroscopy*' [\[ADS link\]](#)

- R. Monier, **D. M. Bowman**, Y. Lebreton, M. Deal, (2023), AJ, Volume 166, Issue 2, 73 ‘*The unexpected optical and ultraviolet variability of the standard star  $\alpha$  Sex (HD 87887)*’ [\[ADS link\]](#)
- R. Monier, E. Niemczura, D. W. Kurtz, S. Rappaport, **D. M. Bowman**, S. J. Murphy, Y. Lebreton, R. Stuik, M. Deal, T. Merle, T. Kiliçoğlu, M. Gebran, E. Le Ster, (2023), AJ, Volume 166, Issue 2, 54, ‘*The surface composition of six newly discovered chemically peculiar stars. Comparison to the HgMn stars  $\mu$  Lep and  $\beta$  Scl and the superficially normal B star  $\nu$  Cap*’ [\[ADS link\]](#)
- A. I. Henriksen, V. Antoci, H. Saio, F. Grundahl, H. Kjeldsen, T. Van Reeth, **D. M. Bowman**, P. I. Pápics, P. de Cat, J. Kruger, and the SONG team, (2023), MNRAS, Volume 524, Issue 3, 4196–4211, ‘*Unresolved Rossby and gravity modes in 214 A and F stars showing rotational modulation*’ [\[ADS link\]](#)
- J. S. Vink, A. Mehner, P. A. Crowther, A. Fullerton, M. Garcia, F. Martins, N. Morrell, L. M. Oskinova, N. St-Louis, A. ud-Doula, A.A.C. Sander, H. Sana, J.-C. Bouret, B. Kubátová, P. Marchant, L. P. Martins, A. Wofford, J. Th. van Loon, O. Grace Telford, Y. Götzberg, **D. M. Bowman**, C. Erba, V. M. Kalari, M. Abdul-Masih, T. Alkousa, F. Backs, C. L. Barbosa, S.R. Berlanas, M. Bernini-Peron, J. M. Bestenlehner, R. Blomme, J. Bodensteiner, S. A. Brands, C. J. Evans, A. David-Uraz, F. A. Driessen, K. Dsilva, S. Geen, V. M. A. Gómez-González, L. Grassitelli, W.-R. Hamann, C. Hawcroft, A. Herrero, E. R. Higgins, D. J. Hillier, R. Ignace, A. G. Istrate, L. Kaper, N. D. Kee, C. Kehrig, Z. Keszthelyi, J. Klencki, A. de Koter, R. Kuiper, E. Laplace, C. J. K. Larkin, R. R. Lefever, C. Leitherer, L. Mahy, J. Maíz Apellániz, G. Maravelias, W. Marcolino, A. F. McLeod, S. E. de Mink, F. Najarro, M. S. Oey, T. N. Parsons, D. Pauli, M. G. Pedersen, R.K. Prinja, V. Ramachandran, M. C. Ramírez-Tannus, G. N. Sabhahit, A. Schootemeijer, S. Reyer Serantes, T. Shenar, G. S. Stringfellow, N. Sudnik, F. Tramper, L. Wang, (2023), A&A, Volume 675, A154, ‘*X-Shooting ULLYSES: Massive Stars at low metallicity. I. Survey Description*’ [\[ADS link\]](#)
- N. Vernekar, A. Subramaniam, V. V. Jadhav, **D. M. Bowman**, (2023), MNRAS, Volume 524, Issue 1, 1360–1373, ‘*Photometric variability of blue straggler stars in M67 with TESS and K2*’ [\[ADS link\]](#)
- [S. Burssens](#), **D. M. Bowman**, M. Michielsen, S. Simón-Díaz, C. Aerts, V. Vanlaer, G. Banyard, N. Nardetto, R. H. D. Townsend, G. Handler, J. S. G. Mombarg, R. Vanderspek, G. Ricker, (2023), Nature Astronomy, Volume 7, 913–930, ‘*A calibration point for stellar evolution from massive star asteroseismology*’ [\[ADS link\]](#)
- D. Pauli, L. M. Oskinova, W.-R. Hamann, **D. M. Bowman**, H. Todt, T. Shenar, A. A. C. Sander, C. Erba, V. M. A. Gómez-González, C. Kehrig, J. Klencki, R. Kuiper, A. Mehner, S. E. de Mink, M. S. Oey, V. Ramachandran, A. Schootemeijer, S. Reyer Serantes, A. Wofford, (2023), A&A, Volume 673, A40 ‘*Spectroscopic and evolutionary analyses of the binary system AzV 14 outline paths towards the WR stage at low-metallicity*’ [\[ADS link\]](#)
- T. Van Reeth, C. Johnston, J. Southworth, J. Fuller, **D. M. Bowman**, L. Poniatoski, J. Van Beeck, (2023), A&A, Volume 671, A121 ‘*Tidally perturbed g-mode pulsations in a sample of close eclipsing binaries*’ [\[ADS link\]](#)
- C. Johnston, A. Tkachenko, T. Van Reeth, **D. M. Bowman**, K. Pavlovski, H. Sana, S. Sekaran, (2023), A&A, Volume 670, A167, ‘*Tidal perturbations and geometric effects on the pulsations in the hierarchical triple system U Gru*’ [\[ADS link\]](#)

**2022:** 2 first author and 12 co-author publications

- **D. M. Bowman**, T. Z. Dorn-Wallenstein, (2022), A&A, Volume 668, A134, ‘*Photometric detection of internal gravity waves in upper main-sequence stars. III. Comparison of amplitude spectrum fitting and Gaussian process regression using CELERITE2*’ [\[ADS link\]](#)
- J. Southworth, **D. M. Bowman**, (2022), The Observatory, Volume 142, 161-173, ‘*Rediscussion of eclipsing binaries. Paper X. The pulsating B-type system V1388 Orionis*’ [\[ADS link\]](#)
- J. Tayar, F. D. Moyano, M. Soares-Furtado, A. Escorza, M. Joyce, S. L. Martell, R. A. García, S. N. Breton, S. Mathis, S. Mathur, V. Delsanti, S. Kiefer, S. Reffert, **D. M. Bowman**, T. Van Reeth, S. Shetye, C. Gehan,

- S. K. Grunblatt, (2022), ApJ, Volume 940, Issue 1, 23, ‘*Spinning up the Surface: Evidence for Planetary Engulfment or Unexpected Angular Momentum Transport?*’ [\[ADS link\]](#)
- O. Kobzar, V. Khalack, D. Bohlender, G. Mathys, M. Shultz, **D. M. Bowman**, E. Paunzen, C. Lovekin, A. David-Uraz, J. Sikora, P. Lampens, O. Richard, (2022), MNRAS, Volume 517, Issue 4, 5340–5357, ‘*Analysis of eight magnetic chemically peculiar stars with rotational modulation*’ [\[ADS link\]](#)
  - S. Gebruers, A. Tkachenko, **D. M. Bowman**, T. Van Reeth, S. Burssens, L. IJspeert, L. Mahy, I. Straumit, M. Xiang, H.-W. Rix, C. Aerts, (2022), A&A, Volume 665, A36, ‘*Analysis of high-resolution FEROS spectroscopy for a sample of variable B-type stars assembled from TESS photometry*’ [\[ADS link\]](#)
  - L. Mahy, H. Sana, T. Shenar, M. Abdul-Masih, G. Banyard, J. Bodensteiner, **D. M. Bowman**, K. Dsilva, M. Fabry, C. Hawcroft, N. Langer, P. Marchant, T. Van Reeth, C. Eldridge, (2022), A&A, Volume 664 A159, ‘*Identifying quiescent compact objects in massive Galactic single-lined spectroscopic binaries*’ [\[ADS link\]](#)
  - Z. T. Spetsieri, P. Boumis, A. Chiotellis, S. Akas, S. Derlopa, S. Shetye, D. M.-A. Meyer, **D. M. Bowman**, V. V. Gvaramadze, (2022), MNRAS, Volume 515, Issue 1, 1544–1556, ‘*Discovery of an optical cocoon tail behind the runaway HD 185806*’ [\[ADS link\]](#)
  - J. A. Toalá, **D. M. Bowman**, T. Van Reeth, H. Todt, K. Dsilva, T. Shenar, G. Koenigsberger, S. Estrada-Dorado, L. M. Oskinova, W.-R. Hamann, (2022), MNRAS, Volume 514, Issue 1, 2269–2277, ‘*Multiple variability time-scales of the early nitrogen-rich Wolf-Rayet star WR7*’ [\[ADS link\]](#)
  - J. Southworth, **D. M. Bowman**, (2022), MNRAS, Volume 513, Issue 3, pp. 3191–3209, ‘*High-mass pulsators in eclipsing binaries observed using TESS*’ [\[ADS link\]](#)
  - T. Van Reeth, J. Southworth, J. Van Beeck, **D. M. Bowman**, (2022), A&A, Volume 659, A177, ‘*V456 Cyg: an eclipsing binary with tidally perturbed g-mode pulsations*’ [\[ADS link\]](#)
  - D. Lecoanet, **D. M. Bowman**, T. Van Reeth, (2022), MNRAS Letters, Volume 512, Issue 1, L16–L20, ‘*Asteroseismic inference of the near-core magnetic field strength in the main-sequence B star HD 43317*’ [\[ADS link\]](#)
  - **D. M. Bowman**, B. Vandenbussche, H. Sana, A. Tkachenko, G. Raskin, T. Delabie, B. Vandoren, P. Royer, S. Garcia, T. Van Reeth, and the CubeSpec collaboration, (2022), A&A, Volume 658, A96, ‘*The CubeSpec space mission. I. Asteroseismology of massive stars from time-series optical spectroscopy: Science requirements and target list prioritisation*’ [\[ADS link\]](#)
  - K. Pavlovski, C. A. Hummel, A. Tkachenko, A. Dervişoğlu, C. Kayhan, R. T. Zavala, D. J. Hutter, C. Tycner, T. Şahin, J. Audenaert, R. Baeyens, J. Bodensteiner, **D. M. Bowman**, S. Gebruers, N. E. Jannsen, J. S. G. Mombarg, (2022), A&A, Volume 658, A92, ‘*Dynamical parallax, physical parameters and evolutionary status of the components of the bright eclipsing binary  $\alpha$  Draconis*’ [\[ADS link\]](#)
  - A. Elliott, N. D. Richardson, H. Pablo, A. F. J. Moffat, **D. M. Bowman**, N. Ibrahim, G. Handler, C. Lovekin, A. Popowicz, N. St-Louis, G. A. Wade, K. Zwintz, (2022), MNRAS, Volume 509, Issue 3, 4246–4255, ‘*Five years of BRITE-Constellation photometry of the prototypical luminous blue variable P Cygni: constraining the stochastic low-frequency variability*’ [\[ADS link\]](#)

**2021:** 2 first author and 12 co-author publications

- **D. M. Bowman** and M. Michielsen, (2021), A&A, Volume 656, A158, ‘*Towards a systematic treatment of observational uncertainties in forward asteroseismic modelling of gravity-mode pulsators*’ [\[ADS link\]](#)
- J. Van Beeck, **D. M. Bowman**, M. G. Pedersen, T. Van Reeth, T. Van Hoolst, C. Aerts, (2021), A&A, Volume 655, A59, ‘*Detection of non-linear resonances among gravity modes of slowly pulsating B stars: Results from five iterative pre-whitening strategies*’ [\[ADS link\]](#)
- J. Audenaert, J. S. Kuszelewicz, R. Handberg, A. Tkachenko, D. Armstrong, M. Hon, R. Kgoadi, M. N. Lund, K. J. Bell, L. Bugnet, **D. M. Bowman**, C. Johnston, R. A. García, D. Stello, L. Molnár, E. Plachy,

- D. Buzasi, C. Aerts, and the T'DA collaboration, (2021), *AJ*, Volume 162, Issue 5, 209, 'TESS Data for Asteroseismology (T'DA) Stellar Variability Classification Pipeline: Set-Up and Application to the Kepler Q9 Data' [\[ADS link\]](#)
- D. L. Holdsworth, M. S. Cunha, D. W. Kurtz, V. Antoci, D. R. Hey, **D. M. Bowman**, O. Kobzar, D. L. Buzasi, O. Kochukhov, E. Niemczura, D. Ozuyar, F. Shi, R. Szabó, A. Samadi-Ghadim, Zs. Bognár, L. Fox-Machado, V. Khalack, M. Lares-Martiz, C. C. Lovekin, P. Mikołajczyk, D. Mkrtichian, J. Pascual-Granado, E. Paunzen, T. Richey-Yowell, Á. Sódor, J. Sikora, T. Z. Yang, E. Brunsden, A. David-Uraz, A. Derekas, A. García Hernández, J. A. Guzik, N. Hatamkhani, R. Handberg, T. S. Lambert, P. Lampens, S. J. Murphy, R. Monier, K. R. Pollard, P. Quiral-Manosalva, A. Ramón-Ballesta, B. Smalley, I. Stateva, R. Vanderspek, (2021), *MNRAS*, Volume 506, Issue 1, 1073–1110, 'TESS Cycle 1 observations of *roAp* stars with 2-min cadence data' [\[ADS link\]](#)
  - A. David-Uraz, M. E. Shultz, V. Petit, **D. M. Bowman**, C. Erba, R. A. Fine, C. Neiner, H. Pablo, J. Sikora, A. ud-Doula, G. A. Wade, (2021), *MNRAS*, Volume 504, Issue 4, 4841–4849, 'MOBSTER – IV. Detection of a new magnetic B-type star from follow-up spectropolarimetric observations of photometrically selected candidates' [\[ADS link\]](#)
  - **D. M. Bowman**, J. Hermans, J. Daszyńska-Daszkiewicz, D. L. Holdsworth, A. Tkachenko, S. J. Murphy, B. Smalley, D. W. Kurtz, (2021), *MNRAS*, Volume 504, Issue 3, 4039–4053 'KIC 5950759: a high-amplitude  $\delta$  Sct star with amplitude and frequency modulation near the terminal age main sequence' [\[ADS link\]](#)
  - W. W. Weiss, K. Zwintz, R. Kuschnig, G. Handler, A. F. W. Moffat, D. Baade, **D. M. Bowman**, T. Granzer, T. Kallinger, O. F. Koudelka, C. Lovekin, C. Neiner, H. Pablo, A. Pigulski, A. Popowicz, T. Ramiaramanantsoa, S. Rucinski, K. Strassmeier, G. Wade, (2021), *Universe*, Volume 7, 199, 'Space Photometry with BRITe-Constellation' [\[ADS link\]](#)
  - M. Michielsen, C. Aerts, **D. M. Bowman**, (2021), *A&A*, Volume 650, A175, 'Probing the temperature gradient in the core boundary layer of stars with gravito-inertial modes: the case of KIC 7760680' [\[ADS link\]](#)
  - S. Gebruers, I. Straumit, A. Tkachenko, J. S. G. Mombarg, M. G. Pedersen, T. Van Reeth, G. Li, P. Lampens, A. Escorza, **D. M. Bowman**, P. De Cat, L. Vermeulen, Y. Frémat, J. Bodensteiner, H.-W. Rix, C. Aerts, (2021), *A&A*, Volume 650, A151, 'A homogeneous spectroscopic analysis of a Kepler legacy sample of dwarfs for gravity-mode asteroseismology' [\[ADS link\]](#)
  - T. Shenar, H. Sana, P. Marchant, B. Pablo, N. Richardson, A. F. J. Moffat, T. Van Reeth, R. H. Barbá, **D. M. Bowman**, P. Broos, P. A. Crowther, S. Clark, A. de Koter, S. E. de Mink, K. Dsilva, G. Gräfener, I. D. Howarth, N. Langer, L. Mahy, J. Máiz Apellániz, A. M. Pollock, F. R. N. Schneider, L. Townsley, J. S. Vink, (2021), *A&A*, Volume 650, A147, 'The Tarantula Massive Binary Monitoring V. R144 – a wind-eclipsing binary with a total mass  $\geq 140 M_{\odot}$ ' [\[ADS link\]](#)
  - C. Johnston, N. Aimar, M. Abdul-Masih, **D. M. Bowman**, T. White, C. Hawcroft, H. Sana, S. Sekeran, K. Dsilva, A. Tkachenko, C. Aerts, (2021), *MNRAS*, Volume 503, Issue 1, 1124–1137, 'Characterization of the variability in the O+B eclipsing binary HD 165246' [\[ADS link\]](#)
  - J. Southworth, **D. M. Bowman**, K. Pavlovski, (2021), *MNRAS Letters*, Volume 501, Issue 1, L65–L70, 'A beta Cephei pulsator and a changing orbital inclination in the high-mass eclipsing binary system VV Orionis' [\[ADS link\]](#)
  - M. G. Pedersen, C. Aerts, P. I. Pápics, M. Michielsen, S. Gebruers, T. M. Rogers, G. Molenberghs, S. Burssens, S. García, **D. M. Bowman**, (2021), *Nature Astronomy*, Volume 5, 715–722, 'Internal mixing of rotating stars inferred from dipole gravity modes' [\[ADS link\]](#)
  - T. Steindl, K. Zwintz, **D. M. Bowman**, (2021), *A&A*, Volume 645, A119, 'Tidally perturbed pulsations in the pre-main sequence  $\delta$  Scuti binary RS Cha' [\[ADS link\]](#)

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**2020:** 2 first author and 10 co-author publications

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- S. Sekaran, A. Tkachenko, M. Abdul-Masih, A. Prša, C. Johnston, D. Huber, S. J. Murphy, G. Banyard, A. W. Howard, H. Isaacson, **D. M. Bowman**, C. Aerts, (2020), A&A, Volume 643, A162, '*Tango of celestial dancers: A sample of detached eclipsing binary systems containing g-mode pulsating components. A case study of KIC9850387*' [\[ADS link\]](#)
- **D. M. Bowman**, (2020), Frontiers in Astronomy and Space Sciences, Volume 7, 70, '*Asteroseismology of high-mass stars: new insights of stellar interiors with space telescopes*' (**Invited review**) [\[ADS link\]](#)
- J. Southworth, **D. M. Bowman**, A. Tkachenko, K. Pavlovski, (2020), MNRAS Letters, Volume 497, Issue 1, L19–L23, '*Discovery of  $\beta$  Cep pulsations in the eclipsing binary V453 Cygni*' [\[ADS link\]](#)
- J. Bodensteiner, T. Shenar, L. Mahy, M. Fabry, P. Marchant, M. Abdul-Masih, G. Banyard, **D. M. Bowman**, K. Dsilva, A. J. Frost, C. Hawcroft, M. Reggiani, H. Sana, (2020), A&A, Volume 641, A43, '*Is HR 6819 a triple system containing a black hole? An alternative explanation*' [\[ADS link\]](#)
- L. Horst, P. V. F. Edelmann, R. Andrásy, F. K. Röpké, **D. M. Bowman**, C. Aerts, R. P. Ratnasingam, (2020), A&A, Volume 641, A18, '*Fully compressible simulations of waves and core convection in main-sequence stars*' [\[ADS link\]](#)
- **D. M. Bowman**, S. Burssens, S. Simón-Díaz, P. V. F. Edelmann, T. M. Rogers, L. Horst, F. K. Röpké, C. Aerts, (2020), A&A, Volume 640, A36, '*Photometric detection of internal gravity waves in upper main-sequence stars. II. Combined TESS photometry and high-resolution spectroscopy*' [\[ADS link\]](#)
- T. Shenar, J. Bodensteiner, M. Abdul-Masih, M. Fabry, L. Mahy, P. Marchant, G. Banyard, **D. M. Bowman**, K. Dsilva, C. Hawcroft, M. Reggiani, H. Sana, (2020), A&A Letters, Volume 639, L6, '*The “hidden” companion in LB-1 unveiled by spectral disentangling*' [\[ADS link\]](#)
- S. Burssens, S. Simón-Díaz, **D. M. Bowman**, G. Holgado, M. Michielsen, A. de Burgos, N. Castro, R. H. Barbá, C. Aerts, (2020), A&A, Volume 639, A81, '*Variability of OB stars from TESS southern Sectors 1-13 and high-resolution IACOB and OWN spectroscopy*' [\[ADS link\]](#)
- J. Van Beeck, V. Prat, T. Van Reeth, S. Mathis, **D. M. Bowman**, C. Aerts, (2020), A&A, Volume 638, A149, '*Detecting axisymmetric magnetic fields using gravity modes in intermediate-mass stars*' [\[ADS link\]](#)
- A. Tkachenko, K. Pavlovski, C. Johnston, C. Aerts, M. G. Pedersen, M. Michielsen, **D. M. Bowman**, J. Southworth, V. Tsymbal, (2020), A&A, Volume 637, A60, '*The mass discrepancy in intermediate- and high-mass eclipsing binaries: The need for higher convective core masses*' [\[ADS link\]](#)
- M. Abdul-Masih, G. Banyard, J. Bodensteiner, E. Bordier, **D. M. Bowman**, K. Dsilva, M. Fabry, C. Hawcroft, L. Mahy, P. Marchant, G. Raskin, M. Reggiani, T. Shenar, A. Tkachenko, H. Van Winckel, L. Vermeylen, H. Sana, (2020), Nature, Volume 580, Issue 7805, E11–E15, '*On the signature of a 70-solar-mass black hole in LB-1*' [\[ADS link\]](#)
- V. Prat, S. Mathis, C. Neiner, J. Van Beeck, **D. M. Bowman**, C. Aerts, (2020), A&A, Volume 636, A100, '*Period spacings of gravity modes in rapidly rotating magnetic stars. II. The case of an oblique dipolar fossil magnetic field*' [\[ADS link\]](#)

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**2019:** 4 first author and 14 co-author publications

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- V. Antoci, M. Cunha, **D. M. Bowman**, S. J. Murphy, D. W. Kurtz, T. R. Bedding, C. Borre, S. Christophe, J. Daszyńska-Daszkiewicz, L. Fox-Machado, A. García Hernández, H. Ghasemi, R. Handberg, H. Hansen, A. Hasanzadeh, G. Houdek, C. Johnston, A. B. Justesen, F. Kahraman Alicavus, F. Kotysz, D. Latham, J. Matthews, J. Mønster, E. Niemczura, E. Paunzen, J. P. Sanchez Arias, A. Pigulski, J. Pepper, T. Richey-Yowell, H. Safari, S. Seager, B. Smalley, T. Shutt, A. Sódor, J.-C. Suárez, A. Tkachenko, T. Wu, K. Zwintz, S. Barceló Forteza, E. Brunsden, Z. Bognár, D. Buzasi, S. Chowdhury, P. De Cat, J. Evans, Z. Guo, J. A. Guzik, N. Jevtic, P. Lampens, M. Lares Martiz, C. Lovekin, G. Li, G. M. Mirouh, D. Mkrtichian, M. J. P. F. G. Monteiro, J. Nemec, R. Ouazzani, J. Pascual-Granado, D. Reese, M. Rieutord, J. R. Rodon, M. Skarka,

- P. Sowicka, I. Stateva, R. Szabó, W. W. Weiss, (2019), MNRAS, Volume 490, Issue 3, 4040–4059, ‘*The first view of  $\delta$  Sct and  $\gamma$  Dor stars with the TESS mission*’ [\[ADS link\]](#)
- V. Khalack, C. Lovekin, **D. M. Bowman**, O. Kobzar, A. David-Uraz, E. Paunzen, J. Sikora, P. Lenz, O. Kochukhov, D. L. Holdsworth, G. A. Wade, (2019), MNRAS, Volume 490, Issue 2, 2102–2111, ‘*Rotational and pulsational variability in the TESS light curve of HD 27463*’ [\[ADS link\]](#)
  - [S. Burssens](#), **D. M. Bowman**, C. Aerts, M. G. Pedersen, E. Moravveji, B. Buysschaert, (2019), MNRAS, Volume 489, Issue 1, 1304–1320, ‘*New  $\beta$  Cep pulsators discovered with K2 space photometry*’ [\[ADS link\]](#)
  - B. J. S. Pope, G. R. Davies, K. Hawkins, T. R. White, A. Stokholm, A. Bieryla, D. W. Latham, M. Lucey, C. Aerts, S. Aigrain, V. Antoci, T. R. Bedding, **D. M. Bowman**, A. Chontos, G. A. Esquerdo, D. Huber, P. Jofré, S. J. Murphy, T. Van Reeth, V. Silva Aguirre, J. Yu, (2019), ApJS, Volume 244, Issue 1, 18, ‘*The Kepler Smear Campaign: Light curves for 102 Very Bright Stars*’ [\[ADS link\]](#)
  - **D. M. Bowman**, C. Johnston, A. Tkachenko, D. Mkrtichian, K. Gunsriwawat, C. Aerts, (2019), ApJ Letters, Volume 883, Issue 1, L26, ‘*Discovery of tidally-perturbed pulsations in the eclipsing binary system U Gru: a crucial system for tidal asteroseismology*’ [\[ADS link\]](#)
  - **D. M. Bowman** and D. L. Holdsworth, (2019), A&A, Volume 629, A21, ‘*Adaptive elliptical aperture photometry: a software package for high-cadence ground-based photometry. I. Application to rapid oscillators observed from SAAO*’ [\[ADS link\]](#)
  - J. Sikora, A. David-Uraz, S. Chowdhury, **D. M. Bowman**, G. A. Wade, V. Khalack, O. Kobzar, O. Kochukhov, C. Neiner, E. Paunzen, (2019), MNRAS, Volume 487, Issue 4, 4695–4710, ‘*MOBSTER – II. Identification of rotationally variable A stars observed with TESS in sectors 1–4*’ [\[ADS link\]](#)
  - M. S. Cunha, V. Antoci, D. L. Holdsworth, D. W. Kurtz, L. A. Balona, Zs. Bognár, **D. M. Bowman**, Z. Guo, P. P. A. Kolaczek-Szymański, M. Lares-Martiz, E. Paunzen, M. Skarka, B. Smalley, Á. Sódor, O. Kochukhov, T. R. Bedding, D. L. Buzasi, L. Fox-Machado, A. Hasanzadeh, E. Niemczura, P. Quiral-Manosalva, I. Stateva, P. De Cat, A. García Hernández, H. Ghasemi, G. Handler, J. M. Matthews, M. J. P. F. G. Monteiro, J. M. Nemec, J. Pascual-Granado, H. Safari, J. C. Suárez, R. Szabó, A. Tkachenko, W. W. Weiss, (2019), MNRAS, Volume 487, Issue 3, 3523–3549, ‘*Rotation and pulsation in Ap stars: first light results from TESS sectors 1 and 2*’ [\[ADS link\]](#)
  - R. Manick, D. Kamath, H. Van Winkel, A. Jorissen, S. Sekaran, **D. M. Bowman**, G.-M. Oomen, J. Kluska, D. Bollen, C. Waelkens, (2019), A&A, Volume 628, A40, ‘*Spectroscopic binaries RV Tauri and DF Cygni*’ [\[ADS link\]](#)
  - A. David-Uraz, C. Neiner, J. Sikora, **D. M. Bowman**, V. Petit, S. Chowdhury, G. Handler, M. Pergeorelis, M. Cantiello, C. Erba, Z. Keszthelyi, V. Khalack, O. Kobzar, O. Kochukhov, J. Labadie-Bartz, R. MacInnis, S. P. Owocki, H. Pablo, M. E. Shultz, A. ud-Doula, G. A. Wade, and the MOBSTER Collaboration, (2019), MNRAS, Volume 487, Issue 1, 304–317, ‘*Magnetic OB[A] stars with TESS: probing their evolutionary and rotational properties (MOBSTER) - I. First-light observations of known magnetic B and A stars*’ [\[ADS link\]](#)
  - V. Prat, S. Mathis, B. Buysschaert, J. Van Beeck, **D. M. Bowman**, C. Aerts, C. Neiner, (2019), A&A, Volume 627, A64, ‘*Period spacings of gravity modes in rapidly rotating magnetic stars. I. Axisymmetric fossil field with poloidal and toroidal components*’ [\[ADS link\]](#)
  - **D. M. Bowman**, S. Burssens, M. G. Pedersen, C. Johnston, C. Aerts, B. Buysschaert, M. Michielsen, A. Tkachenko, T. M. Rogers, P. V. F. Edelmann, R. P. Ratnasingam, S. Simón-Díaz, N. Casto, E. Moravveji, B. J. S. Pope, T. R. White, P. De Cat, (2019), Nature Astronomy, Volume 3, Number 8, 760–765, ‘*Low-frequency gravity waves in blue supergiants revealed by high-precision space photometry*’ [\[ADS link\]](#)
  - J. S. G. Mombarg, T. Van Reeth, M. G. Pedersen, G. Molenberghs, **D. M. Bowman**, C. Johnston, A. Tkachenko, C. Aerts, (2019), MNRAS, Volume 485, Issue 3, 3248–3263, ‘*Asteroseismic masses, ages and core properties of  $\gamma$  Doradus stars using the asymptotic period spacing and spectroscopy*’ [\[ADS link\]](#)
  - P. V. F. Edelmann, R. P. Ratnasingam, M. G. Pedersen, **D. M. Bowman**, V. Prat, T. M. Rogers, (2019), ApJ, Volume 876, Issue 1, 4–24, ‘*Three-dimensional simulations of massive stars I. wave generation and propagation*’ [\[ADS link\]](#)

- G. Handler, A. Pigulski, J. Daszyńska-Daszkiewicz, A. Irrgang, D. Kilkeny, Z. Guo, N. Przybilla, F. Kahraman Aliçavuş, T. Kallinger, J. Pascual-Granado, E. Niemczura, T. Róžański, S. Chowdhury, D. L. Buzasi, G. M. Mirouh, **D. M. Bowman**, C. Johnston, M. G. Pedersen, S. Simón-Díaz, E. Moravveji, K. Gazeas, P. De Cat, R. K. Vanderspek, G. R. Ricker, (2019), *ApJ Letters*, Volume 873, Issue 1, L4, ‘*Asteroseismology of massive stars with the TESS mission: the runaway  $\beta$  Cep pulsator PHL 346 = HN Aqr*’ [\[ADS link\]](#)
- M. G. Pedersen, S. Chowdhury, C. Johnston, **D. M. Bowman**, C. Aerts, G. Handler, P. De Cat, C. Neiner, A. David-Uraz, D. Buzasi, A. Tkachenko, S. Simón-Díaz, E. Moravveji, J. Sikora, G. Mirouh, C. C. Lovekin, M. Cantiello, J. Daszyńska-Daszkiewicz, A. Pigulski, (2019), *ApJ Letters*, Volume 872, Issue 1, L9, ‘*Diverse variability of O and B stars revealed from 2-minute light curves in sectors 1 and 2 of the TESS mission: selection of an asteroseismic sample*’ [\[ADS link\]](#)
- C. Johnston, A. Tkachenko, C. Aerts, G. Molenberghs, **D. M. Bowman**, M. G. Pedersen, B. Buysschaert, P. I. Pápics, (2019), *MNRAS*, Volume 482, Issue 1, 1231–1246, ‘*Binary Asteroseismic Modelling: isochrone-cloud methodology and application to Kepler gravity mode pulsators*’ [\[ADS link\]](#)
- **D. M. Bowman**, C. Aerts, C. Johnston, M. G. Pedersen, T. M. Rogers, P. V. F. Edelmann, S. Simón-Díaz, T. Van Reeth, B. Buysschaert, A. Tkachenko, S. A. Triana, (2019), *A&A*, Volume 621, A135, ‘*Photometric detection of internal gravity waves in upper main-sequence stars. I. Methodology and application to CoRoT targets*’ [\[ADS link\]](#)

**2018:** 2 first author and 8 co-author publications

- D. L. Holdsworth, M. S. Cunha, H. Shibahashi, D. W. Kurtz, **D. M. Bowman**, (2018), *MNRAS*, Volume 480, Issue 3, 2976–2984, ‘*K2 observations of the rapidly oscillating Ap star 33 Lib (HD 137949): new frequencies and unique non-linear interactions*’ [\[ADS link\]](#)
- D. L. Holdsworth, H. Saio, R. R. Sefako, **D. M. Bowman**, (2018), *MNRAS*, Volume 480, Issue 2, 2405–2410, ‘*LCO observations of a super-critical distorted pulsation in the roAp star J0855 (TYC 2488-1241-1)*’ [\[ADS link\]](#)
- T. Van Reeth, J. S. G. Mombarg, S. Mathis, A. Tkachenko, J. Fuller, **D. M. Bowman**, B. Buysschaert, C. Johnston, A. García Hernández, J. Goldstein, R. H. D. Townsend, C. Aerts, (2018), *A&A*, Volume 618, A24, ‘*Sensitivity of gravito-inertial modes to differential rotation in intermediate-mass main-sequence stars*’ [\[ADS link\]](#)
- B. Buysschaert, C. Neiner, A. J. Martin, C. Aerts, **D. M. Bowman**, M. E. Oksala, T. Van Reeth, (2018), *MNRAS*, Volume 478, Issue 2, 2777–2793, ‘*Detection of magnetic fields in chemically peculiar stars observed with the K2 space mission*’ [\[ADS link\]](#)
- B. Buysschaert, C. Aerts, **D. M. Bowman**, C. Johnston, T. Van Reeth, M. G. Pedersen, C. Neiner, (2018), *A&A*, Volume 616, A77, ‘*Forward seismic modeling of the pulsating magnetic B-type star HD 43317*’ [\[ADS link\]](#)
- **D. M. Bowman**, B. Buysschaert, C. Neiner, P. I. Pápics, M. E. Oksala, C. Aerts, (2018), *A&A*, Volume 616, A77, ‘*K2 space photometry reveals rotational modulation and stellar pulsations in chemically peculiar A and B stars*’ [\[ADS link\]](#)
- C. Aerts, G. Molenberghs, M. Michielsen, M. G. Pedersen, R. Björklund, C. Johnston, J. S. G. Mombarg, **D. M. Bowman**, B. Buysschaert, P. I. Pápics, S. Sekaran, J. O. Sundqvist, A. Tkachenko, K. Truyaert, T. Van Reeth, E. Vermeyen, (2018), *ApJS*, Volume 237, 15–46, ‘*Forward asteroseismic modeling of stars with a convective core from gravity-mode oscillations: parameter estimation and stellar model selection*’ [\[ADS link\]](#)
- **D. M. Bowman** and D. W. Kurtz, (2018), *MNRAS*, Volume 476, Issue 3, 3169–3184, ‘*Characterizing the observational properties of  $\delta$  Sct stars in the era of space photometry from the Kepler mission*’ [\[ADS link\]](#)
- C. Aerts, **D. M. Bowman**, S. Simón-Díaz, B. Buysschaert, C. Johnston, E. Moravveji, P. G. Beck, P. De Cat, S. Triana, S. Aigrain, N. Castro, D. Huber, T. White, (2018), *MNRAS*, Volume 476, Issue 1, 1234–1241,

*'K2 photometry and HERMES spectroscopy of the blue supergiant  $\rho$  Leo: rotational wind modulation and low-frequency waves'* [\[ADS link\]](#)

- D. L. Holdsworth, H. Saio, **D. M. Bowman**, D. W. Kurtz, R. R. Sefako, M. Joyce, T. Lambert, B. Smalley, (2018), MNRAS, Volume 476, Issue 1, 601–616, *'Suppressed phase variations in a high amplitude rapidly oscillating Ap star pulsating in a distorted quadrupole mode'* [\[ADS link\]](#)

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**2016:** 1 first author and 1 co-author publications

- **D. M. Bowman**, D. W. Kurtz, M. Breger, S. J. Murphy, D. L. Holdsworth, (2016), MNRAS, Volume 460, Issue 2, 1970–1989, *'Amplitude modulation in  $\delta$  Sct stars: statistics from an ensemble study of Kepler targets'* [\[ADS link\]](#)
- D. W. Kurtz, **D. M. Bowman**, S. J. Ebo, P. Moskalik, R. Handberg, M. N. Lund, (2016), MNRAS, Volume 455, Issue 2, 1237–1245, *'EPIC 201585823, a rare triple-mode RR Lyrae star discovered in K2 mission data'* [\[ADS link\]](#)

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**2015:** 1 first author and 2 co-author publications

- D. W. Kurtz, H. Shibahashi, S. J. Murphy, T. R. Bedding, **D. M. Bowman**, (2015), MNRAS, Volume 450, Issue 3, 3015–3029, *'A unifying explanation of complex frequency spectra of  $\gamma$  Dor, SPB and Be stars: combination frequencies and highly non-sinusoidal light curves'* [\[ADS link\]](#)
- E. Niemczura, S. J. Murphy, B. Smalley, K. Uytterhoeven, A. Pigulski, H. Lehmann, **D. M. Bowman**, G. Catanzaro, E. van Aarle, S. Bloemen, M. Briquet, P. De Cat, D. Drobek, L. Eyer, J. F. S. Gameiro, N. Gorlova, K. Kamiński, P. Lampens, P. Marcos-Arenal, P. I. Pápics, B. Vandenbussche, H. Van Winckel, M. Stęślicki, M. Fagas, (2015), MNRAS, Volume 450, Issue 3, 2764–2783, *'Spectroscopic survey of Kepler stars. I. HERMES/Mercator observations of A- and F-type stars'* [\[ADS link\]](#)
- **D. M. Bowman**, D. L. Holdsworth, D. W. Kurtz, (2015), MNRAS, Volume 449, Issue 1, 1004–1010, *'Combining WASP and Kepler data: the case of the  $\delta$  Sct star KIC 7106205'* [\[ADS link\]](#)

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**2014:** 1 first author publication

- **D. M. Bowman** and D. W. Kurtz, (2014), MNRAS, Volume 444, Issue 2, 1909–1918, *'Pulsational frequency and amplitude modulation in the  $\delta$  Sct star KIC 7106205'* [\[ADS link\]](#)

## XV. Monographs and Book Chapters

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- **D. M. Bowman**, (2017), Springer Theses, Springer International Publishing AG (Springer Nature), '*Amplitude Modulation of Pulsation Modes in Delta Scuti Stars*', DOI: [10.1007/978-3-319-66649-5](https://doi.org/10.1007/978-3-319-66649-5), ISBN: 978-3-319-66648-8

## XVI. Thesis

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- **D. M. Bowman**, (2016), PhD Thesis, Jeremiah Horrocks Institute, University of Lancashire, UK, '*Amplitude modulation and energy conservation of pulsation modes in delta Scuti stars*', online at: <http://clock.uclan.ac.uk/18788/>

## XVII. Conference Proceedings

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- A. J. Frost, H. Sana, L. Mahy, G. Wade, J. Barron, J.-B. Le Bouquin, A. Mérand, F. R. N. Schneider, T. Shenar, R. H. Barbá, **D. M. Bowman**, M. Fabry, A. Farhang, P. Marchant, N. I. Morrell, J. V. Smoker, (2026), Astronomy in Focus as presented at the IAU XXXII General Assembly, 2024, Proceedings of the International Astronomical Union, Volume 20, Issue A32, 167-169, '*An interaction creates an enriched nebula around a massive magnetic binary system*', [\[ADS link\]](#)
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## XVIII. Varia Other Publications

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- E. Šipková, A. Kemp, D. Fritzewski, A. Tkachenko **D. M. Bowman**, C. Aerts, J. Vrancken, (submitted, 2025), *ESO white paper in response to ESO's 'Expanding Horizons' call, 'Exploiting tidal asteroseismology in binary populations from combined space photometry and time-resolved high-resolution spectroscopy'* [[ADS link](#)]

- P. Maxted, **D. M. Bowman**, T. G. Wilson, S. Rosu, K. G. Stassun, S. J. Murphy, A. Moharana, A. H. M. J. Triaud, A. Hahlin, J. Southworth, (submitted, 2025), *ESO white paper in response to ESO's 'Expanding Horizons' call, 'Detached eclipsing binary star science in the 2040s'* [\[ADS link\]](#)
- **D. M. Bowman**, (2024), *Astronomy & Geophysics*, Volume 65, Issue 2, 2.20–2.25, *'Pulsating massive stars'* [\[ADS link\]](#)
- V. Mainieri, R. I. Anderson, J. Brinchmann, *et al.* (total of over 200 co-authors, including **D. M. Bowman**, in the WST consortium), (2024), *'The Wide-field Spectroscopic Telescope (WST) Science White Paper'* [\[ADS link\]](#)
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